NESTED-SPACE COSMOLOGY

The Geometry of a Universe

Regions, boundaries and what an interior observer can learn

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Abstract

Nested-Space Cosmology (NSC) investigates whether one inherited geometric law can connect nested regions, their exchange of energy across event horizons, and the quantities measured by an interior observer. The foundation rests on five openly declared geometric axioms (**A0--A4**): (**A0**) geometry follows the path of least resistance by minimizing the 1D Cauchy--Schwarz generator stretch cost plus enclosing compression work, $E = Q_{stretch} + PV = \sum_{i=1}^{3} m_i/\lambda_i + PV_0\lambda_1\lambda_2\lambda_3$, which arises directly as the 1D vortex-bundle ($A_{\perp_i}=\pi/\mu^2(s_*)=const$) and quantized cavity-mode reduction of the covariant 3+1D field Hamiltonian $\int_V (T^{00}_{phase} + V(s_*))\, d^3x$ with background vacuum pressure $P = V(s_*)$ (Proposition 1.1), while natural boundary variation of the covariant action across the smooth `0` horizon interface Σ_0 enforces chemical-potential equilibrium $\mu_+ = \mu_- \Leftrightarrow \partial E/\partial\theta = 0$ (Proposition 1.1b); (**A1**) a 1D spatial seed plus an inherited compression offset (time) generates a closed 2D circle and 3D sphere; (**A2**) seen edge-on, the closed circle $z(\tau)=R_0e^{i\omega\tau}$ is the harmonic pendulum $R(\tau)=R_0cos(\omega\tau)$, which crosses `0` smoothly into the negative direction $R<0$ (`-`), flipping the 3D orientation Jacobian $det\ F=(R/R_0)^3$ from `+` to `-` and turning inward implosion ($\dot{R}<0$) into outward child expansion ($\dot{r}_- = -\dot{R} > 0$) without the artificial velocity cusp $\Delta\dot{r}=+2R_0\omega$ created when radius is restricted to an unsigned size $r=|R|\geq 0$, while in 3+1D curved geometry the regular bounce areal radius $R(R)=\sqrt{(R^2+R_{core}^2)}$ with standing-wave hard core $R_{core}=\sqrt{2\pi/\kappa}$ and hybrid radial inertia $M(R)=1/(1+R^2/3)$ bounds the Kretschmann scalar at $K(0)=(3\kappa^2)/(\pi^2)(1+R_0^2\omega^2)^2<\infty$ and matches the orthonormal Einstein tensor $G_{\hat{a}\hat{b}}(0)=\kappa T_{\hat{a}\hat{b}}(0)=(1)/(R_{core}^2)diag(-1,-1,+1,+1)$ directly to the `(1,1,1)` standing-wave angular floor and radial Casimir winding tension (Propositions 1.2--1.2b); (**A3**) each nested room carries a moving wavelength band $[\lambda_{min}(\chi),\lambda_{max}(\chi)]=e^{-\chi}[\lambda_{min,0},\lambda_{max,0}]$ bounded by its Planck cutoff (λ_{min}) and Absolute-Zero horizon (λ_{max}); and (**A4**) every room sits between an outer birth `0` valve and inner black-hole `0` valves on a closed vortex--diffusion ring (`0 <-> 0`).

Minimizing the unified 5-variable single-cost functional across `+`, `0`, and `-` yields a unique closed-form global minimum (Section 9) and seven parameter-free geometric, dynamical, and observational theorems:

- **(1) Exact algebraic budget surface and open-valve Friedmann equations of state:** Eliminating both internal variables (θ^*,χ) from the five-variable budget shares (D,M,V) with $B=M-D/3$ yields the exact zero-parameter algebraic curve $9DVB=4(V+B/2)^3$, exact inverse recovery $\theta^*=D/(D+V+B/2)$ and $e^{-\chi}=3\sqrt{\theta^*}B/(2V+B)$ (equivalently $e^{2\chi}=V(D+V+B/2)/[B(V+B/2)]$), and the deep-nest invariant $lim_{\chi\to\infty} B/V^2=16/27$. Along the open-valve Virial manifold $\theta^*(a_-)=(PV_0/m_0)^{1/2}a_-^2$ ($V_-=V_0\,a_-^3$), the total differential $dE_{DE}^*=(-(1)/(3)\rho_{DE}+(4)/(3)\rho_{DE})dV_-=+\rho_{DE}\,dV_-$ yields $w_{DE}=-(dE_{DE}^*/dV_-)/\rho_{DE}=-1$ ($\rho_{DE}=3P=const$), while the boundary-supported horizon load E_{DM}^* does zero interior bulk work ($w_{DM}=0$, $c_{s,DM}^2=0$), matching the covariant massive-scalar late tail $\langle p_\chi\rangle/\langle\rho_\chi\rangle\to 0$.
- **(2) Universal exponential-wall bypasses, `(1,1,1)` 3D triad ground state, and $O(1)$ dilution survival:** Above $\chi_{spaghetti}=(1)/(2)ln\ 2$, the infalling `+` parcel shrinks in volume while spaghettifying into a 1D filament ($\theta^*=1/2$), while three scale-free geometric mechanisms ($d\lambda/\lambda$ logarithmic measure, active vortex-tube circulation $w_{tube}=1$, and zero-mode topological phase $\Delta\theta/\pi=1/2$) bypass the e^{-3K} volumetric dilution wall at $O(1)$. In 3D standing-wave closure, the orthogonal `(1,1,1)` triad is the unique minimum-cost 3D ground state ($E_{1,1,1}=3\pi^2/R^2$, *2:1* split against the S^2 angular floor) with wave-compression hard core $R_{core}=\sqrt{2\pi/\kappa}$ and short-range nuclear hard-core repulsion $\Delta E_{core}=8(\sqrt{2}-1)$ against merging into `(2,2,2)`.
- **(3) Gauss-divergence *3:1* exterior Virial partition, helical buckling, and damped `+/0/-` oscillator:** For any incompressible irrotational potential flow $\varphi(r,\tau)=-R^2\dot{R}/r$ around a spherical throat of radius $R(\tau)$, the Gauss divergence identity $\nabla\cdot(x p_{dyn})=3p_{dyn}+x\cdot\nabla\, p_{dyn}$ decomposes the exterior pressure-gradient Virial $\int_{r\geq R} x\cdot(-\nabla\, p_{dyn})\, d^3x=-4T_{ext}$ into the exact *3:1* bulk-to-throat partition $|W_{bulk}|:|W_{throat}|=3T_{ext}:T_{ext}=3:1$ (matching the static *3:1* surrounding-flow to cavity-work ratio $E_{surr}:W_{press}=3:1$), while axial compression buckles laterally into a helix and drives a damped `+/0/-` horizon oscillation with exact asymmetries $\eta_{area}=tanh((\pi\gamma)/(2\omega))$ and $\eta_{work}=tanh((\pi\gamma)/(\omega))=2\eta_{area}/(1+\eta_{area}^2)$.

- **(4) Coupled finite-parent / receiver continuation, continuum zero-radius opening, and last-echo cutoff:** Coupling a finite parent reservoir $E_p(\tau)+E_-(\tau)=E_{tot}$ with hybrid radial inertia $M(r)=1/(1+r^2/3)$ admits a continuum wave-driven opening from zero radius $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$, the closed *3:1* relaxation law $R(\tau)/R_\infty=tanh(\sqrt{2}E_0\,\tilde{\tau})$, and a sharp last-echo horizon cutoff ($\Delta E_{post}/E_{tot}=0$, visible interior snapshot *25.63%*, parent curvature backreaction shift *8.98%*).
- **(5) Spherical fragmentation, golden-ratio causal neck onset $p^3-3p-3=0$, and outgoing multipole waveform:** Integrating the outward causal stress balance over the entire exterior $r\in[1,\infty)$ of the harmonic two-center family gives the boundary release law $f_r(1,t)+4f(1,t)=0$ and the parameter-free characteristic cubic $p^3-3p-3=0$, whose unique real unstable growth root in terms of the golden ratio $\varphi=(1+\sqrt{5})/(2)$ is $p_+=\varphi^{2/3}+\varphi^{-2/3}\approx 2.1038034027$, driving a finite-time neck pinch, positive central observer pulse, and analytical pole-sum late-time tail $\varphi_C(T)\propto -1/T'+O((T')^{-2})$.
- **(6) Three-axis orthogonal post-pinch cascade and 2×2 kinetic handoff:** Harmonicity ($\Delta\Phi=0$) forces the directional enclosed-volume derivative trace $\partial_{\zeta_x} v+\partial_{\zeta_y} v+\partial_{\zeta_z} v=0$ to vanish at every stage, driving a 3-axis orthogonal cascade $e_z\to e_x\to e_y$ ($1\to 2\to 4\to 8$ lobes with exact harmonic ratios $-1/2$ and -2), coupled by the exact multipole identity $\partial_{q_2} g_{l,0}|_{q_2=0}=-(l-1)/(2)\partial_{q_1} g_{l,0}$ and a strictly positive-definite, B-independent 2×2 quasistatic kinetic matrix ($det\, M_{full}>0$, initial transverse conjugate momentum $p_2(0)>0$).
- **(7) Topological 3-axis cascade freeze-out ($\chi_{freeze}=3$), WHIM split, $H_0=67.03\ km\ s^{-1}Mpc^{-1}$, and $S_8\approx 0.776$:** Quantizing the scale-invariant logarithmic action $\Delta\chi_k=\int_{e^{-\Delta\chi_k}}^{1} d\lambda_k/\lambda_k=n_k=1$ on the (1,1,1) triad ground state across all $d=3$ mutually orthogonal downhill axes ($e_z\to e_x\to e_y$, consistent with the two-center handoff half-span $L_{handoff}=1.2+2^{2/3}=2.7874\approx e^1$, Lemma 7.1) via **A1** telescoping fixes $\chi_{freeze}=\sum_{k=1}^{3}\Delta\chi_k=3$, selecting $(\Omega_{DE},\Omega_{DM},\Omega_{vis})=(70.88\%,23.81\%,5.30\%)$ at $\chi=3$ and $(71.14\%,23.88\%,4.98\%)$ at $\Omega_{vis}=e^{-3}$ on $9DVB=4(V+B/2)^3$, predicting the tangent drift slope $d\Omega_{DE}/d\Omega_{vis}=-(3)/(4)-(8)/(9)\Omega_{vis}+O(\Omega_{vis}^2)\approx -0.806$, separating condensed (1,1,1) baryon cores ($\Omega_b=e^{-3}=4.9787\%$, fixing $H_0=100\sqrt{(\Omega_b\, h^2\, e^3)}=67.03\pm 0.22\ km\ s^{-1}Mpc^{-1}$) from uncondensed 1D WHIM filaments ($\Delta\Omega_{WHIM}=0.3252\%$), and suppressing sub-horizon linear growth via the spatially uniform horizon-throat fraction ($1-f_{clust}=3.114\%$) from $S_{8,CMB}\approx 0.832$ to $S_{8,NSC}\approx 0.776$ (Proposition 9.7b).

Part I (Sections 2--8) presents the covariant 3+1D two-scalar Einstein control apparatus and the pre-registered single-horizon flow test NSC-P006, proving that an uncoupled single horizon leaves $\Omega_{\Lambda,0}$ unselected and motivating Part II (Section 9), while Section 9.8 strictly quarantines non-derived numerical comparisons and Appendix A provides the complete reproducibility index.

1. One spatial seed, offset history and a generated geometry

Douglas's proposal is that contrast and exchange organize approximately spherical regions, and that the construction can repeat within them. Time names the offset operation in this generating picture. A physical duration is a readout along the resulting history. The generated space has three spatial directions; a child is another region, not a fourth primitive spatial direction.

The circle and sphere separate an inside from an outside. Isotropic positive surface energy at fixed volume selects a sphere by the isoperimetric inequality. Whether a particular gravitational region is selected by that variational problem must follow from its action. The present smooth horizon has no material tension. A material-shell branch is therefore retained separately in the technical atlas.

The explicit two-stage offset generator is

$$\theta_{n+1} = \theta_n + \alpha$$
$$\phi_{n+1} = \phi_n + \theta_n$$

with phases taken modulo one and zero initial phases ($\theta_0=\varphi_0=0$). Hence $\theta_n=n\alpha$ and $\varphi_n=(1)/(2)n(n-1)\alpha$ modulo one. One accumulator feeds the second; no independent second phase is supplied. For irrational

α, Weyl's polynomial equidistribution criterion proves that the joint phase history is dense on the two-phase torus T^2.

Setting $\Theta_n = 2\pi\theta_n$ and $\Phi_n = 2\pi\varphi_n$, the generated unit directions and their inner-product kernel are

$$v_n = (\sin\Theta_n\cos\Phi_n, \sin\Theta_n\sin\Phi_n, \cos\Theta_n)$$
$$C_{nm} = v_n \cdot v_m.$$

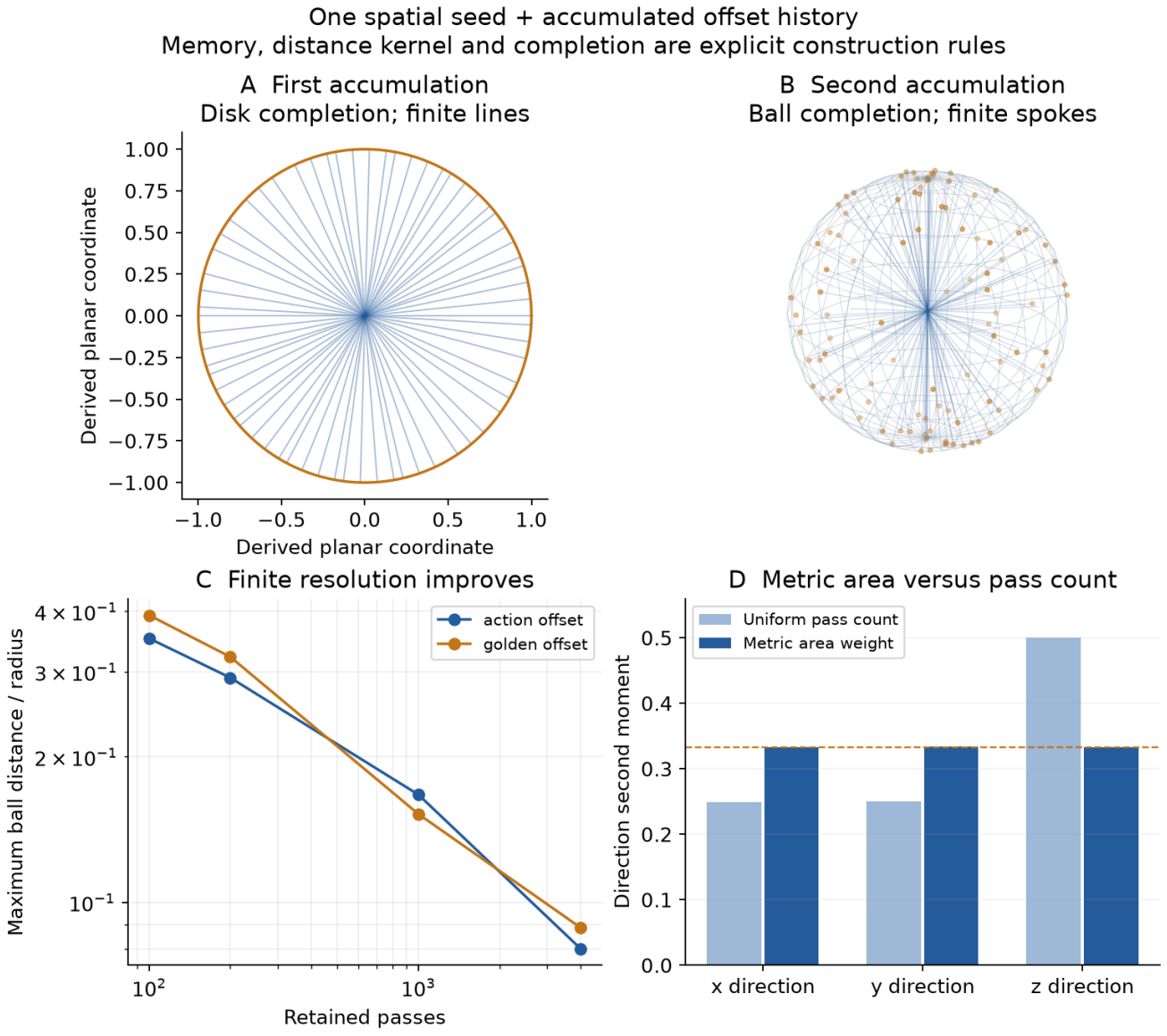
For a retained radial interval, the declared Euclidean distance kernel

$$d^2[(r,n),(s,m)] = r^2 + s^2 - 2rsC_{nm}$$

has the closed three-dimensional ball B^3 as its metric completion. The positive-curvature version

$$d = R\arccos\left[\cos(r/R)\cos(s/R) + \sin(r/R)\sin(s/R)C_{nm}\right]$$

generates the closed spatial parent S^3. Memory, the distance law and metric completion are explicit assumptions; finite drawings have zero three-dimensional volume before completion.



The inherited-law target across nested transitions is

$$\mathcal{T}^{*}\,\mathbb{D}=\mathbb{D}$$

stating that the normalized operator law is preserved under the parent-to-child pullback T^{*}.

1.1 Openly declared geometric axioms (A0--A4) and covariant field reduction

To build a rigorous deductive chain without unstated fluid-physics analogies (such as assuming parcel incompressibility $\lambda_1\lambda_2\lambda_3=1$ or borrowed surface tension), the non-local foundation states five explicit geometric axioms:

- **A0 (Path of Least Resistance):** By Cauchy--Schwarz on any generated 1D phase curve of physical length $L_i=\lambda_i L_{i,0}$ carrying winding change $\Delta\psi_i$, the intrinsic stretch cost is $Q_{stretch}(A)=(1)/(2)Tr[(A^TA)^{-1/2}C^TC]=\sum_{i=1}^{3} m_i/\lambda_i$. Under enclosing compression work PV, geometry minimizes the single cost functional $E=Q_{stretch}+PV$.
- **A1 (1D Seed + Inherited Offset = Time):** Every child seed forms inside an already evolving parent at relative log-length compression $\chi_{p\to c}=(1)/(3)ln[(J_p/J_{p,0})/(J_c/J_{c,0})]$, which satisfies exact telescoping $\chi_{g\to c}=\chi_{g\to p}+\chi_{p\to c}$ across generations. Repeated offset accumulation generates the 2D circle and 3D sphere.
- **A2 (Signed-Zero Crossing + → 0 → -):** The radial coordinate $R(\tau)$ is a signed geometric coordinate rather than an unsigned size. Under winning inward compression $F_{net}=F_{in}-F_{out}>0$, $R(\tau)$ crosses smoothly through `0` into $R<0$ (`-`), flipping the 3D orientation Jacobian $det[(R/R_0)I_3]=(R/R_0)^3$ from `+` to `-` and converting inward implosion ($\dot{R}<0$) into outward child expansion ($\dot{r}_-=-\dot{R}>0$, $r_-=-R>0$).
- **A3 (Moving Spectral Band $[\lambda_{min},\lambda_{max}]$):** Each room at relative compression χ supports a finite wavelength band $[\lambda_{min}(\chi),\lambda_{max}(\chi)]=e^{-\chi}[\lambda_{min,0},\lambda_{max,0}]$ bounded by its shortest wavelength before `0` crosses into `-` (local Planck cutoff λ_{min}) and its longest wavelength as wide as the room itself (local Absolute-Zero horizon $\lambda_{max}=2R$).
- **A4 (Two-Valve Nested Circulation):** Each `0` horizon is a one-way valve from `+` to `-` conserving total turn budget $n_++n_-=N_0$ ($n_+=(1-\theta)N_0$, $n_-=\theta N_0$). Every local room sits between an outer birth `0` valve (inflow from parent) and inner black-hole `0` valves (outflow to children) on a closed vortex--diffusion ring (`0 <-> 0`).

Proposition 1.1 (Reduction of the Covariant 3+1D Field Hamiltonian to A0): Consider the covariant 3+1D scalar-gravity Lagrangian of Section 2, $L=(R)/(2\kappa)-(1)/(2)(\nabla S)^2-(1)/(2)(\nabla\chi)^2-V(s)-(1)/(2)\mu^2(s)\chi^2$, evaluated on an orthorhombic principal domain $V=V_0\lambda_1\lambda_2\lambda_3$ of physical lengths $L_i=\lambda_i L_{i,0}$ ($i=1,2,3$) immersed in the parent background vacuum energy density $P\equiv V(s_*)$. Along each generated 1D phase-curve bundle γ_i of physical length $L_i=\lambda_i L_{i,0}$ and transverse vortex core cross-section $A_{\perp,i}=\pi\, r_\perp^{\,2}=\pi/\mu^2(s_*)$ fixed by the microscopic Compton healing radius $r_\perp=\mu(s_*)^{-1}$ of the stabilized parent scalar mass $\mu^2(s_*)=400e^{-12s_*}$ (which is an intrinsic microscopic property of the scalar potential and therefore strictly independent of the macroscopic domain stretches λ_j), carrying topological phase winding $\Delta\psi_i=\int_0^{L_i}\partial_s\psi_i\, ds=2\pi\, n_i$, the Cauchy--Schwarz inequality on the spatial gradient energy yields

$$\int_{\text{tube}_i}\frac{1}{2}(\partial_s\psi_i)^2\,d^3x=\frac{A_{\perp,i}}{2}\int_0^{L_i}(\partial_s\psi_i)^2\,ds\geq\frac{A_{\perp,i}(\Delta\psi_i)^2}{2L_{i,0}\lambda_i}\equiv\frac{m_i}{\lambda_i}$$

$$m_i=\frac{2\pi^2A_{\perp,i}n_i^2}{L_{i,0}}=\frac{2\pi^3n_i^2}{\mu^2(s_*)L_{i,0}}$$

with equality attained on the harmonic uniform-gradient phase $\partial_s\psi_i=\Delta\psi_i/L_i$ (equivalently, in the quantized relativistic cavity limit at fixed mode occupation N_i, each standing-wave mode of wavenumber $k_i=\pi\, n_i/(\lambda_i\, L_{i,0})$ carries energy $E_i=N_i\hbar\, c\, k_i=(N_i\hbar\, c\pi\, n_i/L_{i,0})/\lambda_i\equiv m_i/\lambda_i$ with m_i strictly independent of

$\lambda_1,\lambda_2,\lambda_3$). Adding the bulk background vacuum energy $\int_V V(s_*)\, d^3x = PV_0\lambda_1\lambda_2\lambda_3$ gives the exact reduced Hamiltonian functional of **A0**:

$$E(\lambda_1,\lambda_2,\lambda_3) = \int_V \left(T^{00}_{\text{phase}} + V(s_*)\right) d^3x = \sum_{i=1}^{3} \frac{m_i}{\lambda_i} + PV_0\lambda_1\lambda_2\lambda_3.$$

Whereas smearing a fixed classical amplitude homogeneously across empty 3D volume dilutes as $\lambda^{d-2}=\lambda^1=e^{-\chi}$, transverse core localization at the Compton healing radius $r_\perp=\mu(s_*)^{-1}$ (or fixed quantum occupation N_i along the generated 1D phase curves) preserves the exact $1/\lambda_i$ Cauchy--Schwarz stretch resistance in every dimension d.

Proposition 1.1b (Covariant 3+1D Junction Derivation of Open-Valve Stationarity $\partial E/\partial\theta = 0$): Let the two-sheeted spacetime be separated into parent (M_+) and child (M_-) domains by the smooth `0` horizon interface Σ_0 ($R=0$, $v=\pi/2$), across which the total topological winding $2\pi N_0$ is partitioned via the horizon boundary phase $\psi_{\Sigma_0} = 2\pi N_0\theta(\tau)$, so that $\Delta\psi_+ = 2\pi N_0(1-\theta)$ on M_+ and $\Delta\psi_- = 2\pi N_0\theta$ on M_- (**Axiom A4**). Evaluating the covariant 3+1D scalar action $S = \int L\sqrt{-g}\, d^4x$ on the harmonic vortex-bundle fields of Proposition 1.1 ($\partial_s\psi_{+,i} = \Delta\psi_+/(\lambda_i L_{i,0})$, $\partial_s\psi_{-,j} = \Delta\psi_-/(\tilde{\lambda}_j L_{j,0})$) and performing the natural boundary variation with respect to the intermediate horizon phase $\delta\psi_{\Sigma_0} = 2\pi N_0\,\delta\theta(\tau)$ yields the canonical junction identity:

$$\delta\mathcal{S}_{\Sigma_0} = \int_{\Sigma_0} d\tau\,(2\pi N_0)\,\delta\theta(\tau)\left[\sum_{i=1}^{3} A_{\perp,i}\,\partial_s\psi_{+,i} - \sum_{j=1}^{3} A_{\perp,j}\,\partial_s\psi_{-,j}\right] = -\int_{\Sigma_0} d\tau\,\frac{\partial E}{\partial\theta}\,\delta\theta(\tau)$$

where $m_0 = 2\pi^2 A_\perp N_0^2/L_0$ and

$$-\frac{\partial E}{\partial\theta} = 2m_0(1-\theta)\sum_{i=1}^{3}\frac{1}{\lambda_i} - 2m_0\theta\sum_{j=1}^{3}\frac{1}{\tilde{\lambda}_j} = 2\pi N_0(\mu_+ - \mu_-)$$

with $\mu_\pm = \partial E_\pm/\partial(2\pi n_\pm)$ denoting the canonical winding chemical potential on each side of Σ_0. Consequently:

- *Closed-valve regime (pinned horizon phase, $\delta\theta = 0$):* When the `0` valve is closed, θ is frozen and adiabatic child expansion $V_- = V_0 a_-^3$ obeys $E_{DE} \propto a_-^{-1}$ ($w_{closed} = +1/3$).
- *Open-valve regime (unpinned smooth horizon, $\delta\theta$ free):* Because the smooth horizon at $v=\pi/2$ has no membrane barrier or singular denominator (Section 2), $\delta S_{\Sigma_0}/\delta\theta = 0$ enforces chemical-potential equilibrium $\mu_+(\chi,\theta) = \mu_-(\theta) \Leftrightarrow \partial E/\partial\theta = 0$ across Σ_0, locking the system onto the open-valve manifold M_{valve} ($\theta^*(a_-) = (PV_0/m_0)^{1/2}a_-^2$) and generating $w_{DE} = -1$ (Section 9.1).

1.2 Circle-pendulum duality: why - exists, `(-) x (-) = (+)`, and curvature regularization at $R=0$

In Layer 1 of the offset generator, the pen draws a closed circle $z(\tau)=R_0e^{i\omega\tau}$. Seen edge-on along any diameter, that uniform circular motion is the harmonic pendulum

$$R(\tau) = \mathrm{Re}\left(R_0e^{i\omega\tau}\right) = R_0\cos(\omega\tau)$$
$$\ddot{R} + \omega^2 R = 0$$

which swings $+R_0 \to 0 \to -R_0 \to 0 \to +R_0$ with invariant total energy $(1)/(2)R^2+(1)/(2)\omega^2\, R^2=(1)/(2)R_0^2\omega^2$. Three exact geometric consequences follow immediately:

- **(1) Why `-` must exist for space to close:** Negative is direction, not less than nothing. If negative coordinates are forbidden ($R \geq 0$ only), the circle is clipped to $\omega\tau \in [-\pi/2, \pi/2]$: exactly *50%* of its arc length (πR_0 out of $2\pi R_0$) is lost and no loop or bounded space can close.
- **(2) Why `(-) x (-) = (+)` is two-generation geometry:** Multiplying by $-1 = e^{i\pi}$ is a geometric half-turn (π rotation). In three spatial dimensions, crossing `0` applies the spatial inversion $P = -I_3$ with $\det P = (-1)^3 = -1$ (*Parent (+)→Child (-)*). Crossing `0` a second time into the grandchild applies two half-turns, $P^2 = (-I_3)(-I_3) = +I_3$ with $\det(P^2) = (-1)(-1) = +1$ (*Child (-)→Grandchild (+)*).
- **(3) Kinematic cusp removal and 3+1D curvature regularization at $R=0$:** Standard physics uses negative coordinates everywhere except for the radius of space itself, which it restricts to an unsigned size $r(\tau) = |R(\tau)| \geq 0$. At the horizon crossing $\tau_0 = \pi/(2\omega)$ ($R=0$), folding the negative half of the swing onto the positive half forces a discontinuous velocity jump $\Delta\dot{r} = \dot{r}(\tau_0^+) - \dot{r}(\tau_0^-) = +2R_0\omega$ and a Dirac-delta acceleration cusp $\ddot{r} = 2R_0\omega\,\delta(\tau - \tau_0) - \omega^2 r$. Keeping the signed coordinate $R(\tau)$ removes this kinematic cusp completely ($\Delta R = 0$, $\ddot{R}(\tau_0) = 0$).

Proposition 1.2 (3+1D Curvature Finiteness Across the Signed-Zero Crossing): In 3+1D curved geometry, combining the signed radial coordinate $R \in (-\infty, \infty)$ with the `(1,1,1)` standing-wave compression hard core $R_{core} = \sqrt{2\pi/\kappa} > 0$ (derived in Theorem 2, Section 9.2) and the hybrid radial inertia metric $M(R) = 1/(1 + R^2/3)$ (derived in Theorem 4, Section 9.4) yields the regular two-sheeted bounce line element

$$
\begin{aligned}
ds^2 &= -d\tau^2 + M(R)\,dR^2 + \mathcal{R}^2(R)\,d\Omega_2^2 \\
\mathcal{R}(R) &\equiv \sqrt{R^2 + R_{\text{core}}^2} \\
M(R) &= \frac{1}{1 + R^2/3}.
\end{aligned}
$$

Because $M(0) = 1$, $R(0) = R_{core} > 0$, $R'(0) = 0$, and $R''(0) = 1/R_{core}$ are analytic at the throat $R=0$, the proper birth distance $\int_0^{\epsilon} \sqrt{M(R)}\,dR < \infty$ is finite and every 3+1D curvature invariant is strictly bounded across the `+ -> 0 -> -` transition:

$$
\begin{aligned}
R_4(0) &= -\frac{2}{R_{\text{core}}^2} = -\frac{\kappa}{\pi} < \infty \\
\mathcal{K}_{\text{static}}(0) &= \frac{12}{R_{\text{core}}^4} = \frac{3\kappa^2}{\pi^2} < \infty \\
\mathcal{K}_{\text{dyn}}(0) &= \frac{3\kappa^2}{\pi^2}\left(1 + R_0^2\omega^2\right)^2 < \infty.
\end{aligned}
$$

Proposition 1.2b (Exact Orthonormal Einstein Tensor and Standing-Wave Stress Matching Across $R=0$): For the two-sheeted bounce line element of Proposition 1.2, the non-vanishing orthonormal sectional curvatures for all $R \in (-\infty, \infty)$ are

$$
\begin{aligned}
R_{\hat{R}\hat{\theta}\hat{R}\hat{\theta}}(R) &= R_{\hat{R}\hat{\phi}\hat{R}\hat{\phi}}(R) = -\frac{R_{\text{core}}^2(1 + R^2/3) + \frac{1}{3}R^2(R^2 + R_{\text{core}}^2)}{(R^2 + R_{\text{core}}^2)^2} \\
R_{\hat{\theta}\hat{\phi}\hat{\theta}\hat{\phi}}(R) &= \frac{R_{\text{core}}^2 - R^4/3}{(R^2 + R_{\text{core}}^2)^2}.
\end{aligned}
$$

The resulting exact orthonormal Einstein tensor components $G_{\hat{a}\hat{b}}(R) = \kappa\, T_{\hat{a}\hat{b}}(R)$ are analytic everywhere on $R \in (-\infty, \infty)$ and satisfy the contracted Bianchi identity $(d)/(dR) G_{\hat{R}\hat{R}} + (2R')/(R)(G_{\hat{R}\hat{R}} - G_{\hat{\theta}\hat{\theta}}) = 0$ identically:

$$G_{\hat{\tau}\hat{\tau}}(R) = -\frac{R^4 + \frac{4}{3}R^2R_{\text{core}}^2 + R_{\text{core}}^2}{(R^2 + R_{\text{core}}^2)^2}$$
$$G_{\hat{R}\hat{R}}(R) = \frac{\frac{1}{3}R^4 - R_{\text{core}}^2}{(R^2 + R_{\text{core}}^2)^2}$$
$$G_{\hat{\theta}\hat{\theta}}(R) = G_{\hat{\phi}\hat{\phi}}(R) = \frac{\frac{1}{3}R^4 + \frac{2}{3}R^2R_{\text{core}}^2 + R_{\text{core}}^2}{(R^2 + R_{\text{core}}^2)^2}.$$

Evaluating this profile at the signed-zero throat $R=0$ and in the asymptotic rooms $R\to\pm\infty$ establishes two exact physical limits:

- **(1) Exact `(1,1,1)` Casimir / standing-wave balance at the throat ($R=0$):** At $R=0$, the required covariant stress-energy tensor is

$$T_{\hat{a}\hat{b}}(0) = \tfrac{1}{\kappa}G_{\hat{a}\hat{b}}(0) = \frac{1}{\kappa R_{\text{core}}^2}\,\text{diag}(-1,\ -1,\ +1,\ +1) = \frac{1}{2\pi}\,\text{diag}(-1,\ -1,\ +1,\ +1).$$

The transverse angular pressure $p_\theta(0)+p_\varphi(0)=2/(\kappa\ R_{core}^{\ 2})$ is supplied directly by the $\ell=1$ S^2 angular kinetic floor $\ell(\ell+1)/R_{core}^{\ 2}=2/R_{core}^{\ 2}$ of the `(1,1,1)` standing-wave triad (Theorem 2(3)), divided equally between $\hat{\theta}$ and $\hat{\varphi}$, while the boost-invariant radial tension $\rho(0)=p_R(0)=-1/(\kappa\ R_{core}^{\ 2})=-1/(2\pi)$ is the 1D radial Casimir winding stress across the throat area $A_{core}=4\pi\ R_{core}^{\ 2}=8\pi^2/\kappa$, providing the exact radial flare-out support $\kappa(\rho(0)+p_R(0))=-2/R_{core}^{\ 2}=-2R''(0)/R(0)<0$.

- **(2) Asymptotic recovery of the *3:1* Virial curvature scale ($R\to\pm\infty$):** For $|R|/R_{core}\to\infty$, the factor $1+R^2/3$ in $M(R)$ drives $G_{\hat{\tau}\hat{\tau}}(\pm\infty)\to -1$ and $G_{\hat{R}\hat{R}}(\pm\infty),G_{\hat{\theta}\hat{\theta}}(\pm\infty)\to +1/3$, smoothly matching the throat core onto a spatial slice of constant curvature radius $L_H=\sqrt{3}$.

2. The smooth zero interface and its inherited source

Architectural overview (Part I: Sections 2--8 vs. Part II: Section 9): To establish mathematically why **Axiom A4 (Two-Valve Nested Circulation)** is necessary, this paper is organized in two complementary parts. **Part I (Sections 2--8)** constructs a fully covariant 3+1D Einstein/two-scalar **single-horizon control apparatus** (`+` parent → `0` horizon → `-` receiver without a second outflow valve or variational child boundary closure). This control apparatus proves three foundational benchmark results: (i) the smooth `0` horizon transmits energy and area growth conservatively without membrane singularities (Sections 2--3); (ii) an isolated single-horizon scalar flux *cannot* select the cosmological vacuum fraction $\Omega_{\Lambda,0}$ on its own (the pre-registered negative theorem NSC-P006, Section 4), because null horizon projections are blind to an additive vacuum shift; and (iii) the transmitted massive scalar phase relaxes asymptotically to a cold, pressureless dust tail with $w_\chi=(p_\chi)/(\rho_\chi)\to 0$ (Section 5). **Part II (Section 9)** then introduces the full **two-valve variational foundation (`A0--A4`)** and proves that closing the circulation loop between the outer birth `0` valve and inner black-hole `0` valves uniquely fixes the algebraic budget surface $9DVB=4(V+B/2)^3$, the Friedmann equations of state $w_{DE}=-1$ and $w_{DM}=0$, and the 3-axis cascade freeze-out $\chi_{freeze}=3$ (Theorems 1--7).

The worked parent uses the Einstein/two-scalar action

$$\mathcal{L} = \frac{R}{2\kappa} - \frac{(\nabla S)^2}{2} - \frac{(\nabla\chi)^2}{2} - W$$
$$W = V(s) + \frac{\mu^2(s)\chi^2}{2}$$

with $S=\sqrt{84}(s-s_*)$ and

$$V(s) = 7e^{-12s} - 7.5e^{-14s} + \frac{25}{32}e^{-24s}$$
$$\mu^2(s) = 400e^{-12s}.$$

Its prepared real scalar contains two spatial harmonics with Laplacian eigenvalues *3* and *8*, equal reference energy, and zero initial velocity. The reference frequency $\omega_0=20$, state amplitude ε, and background curvature H remain declared inputs; the minimum of $V(s)$ supplies the background parent vacuum $P=V(s_*)$.

In conformal null coordinates $u=\eta-\chi$ and $v=\eta+\chi$, the smooth horizon lies at $v=\pi/2$, where the inherited radial fields satisfy

$$4\Psi_{uv} + \left[(m^2-2)\sec^2\eta + 2\csc^2\chi\right]\Psi = 0$$
$$m^2 = \frac{\mu^2}{H^2}.$$

There is no horizon denominator and no membrane stress. Two intersecting null traces inherited from the global preparation fix the receiving field across $v=\pi/2$ without resetting the state.

3. One source, three connected crossing numbers

The source's null stress gives a positive regional Killing-energy flux:

$$\frac{dE_K}{dt} = -\varepsilon^2 \tanh(Ht) F_H(t)$$
$$F_H(t) \geq 0.$$

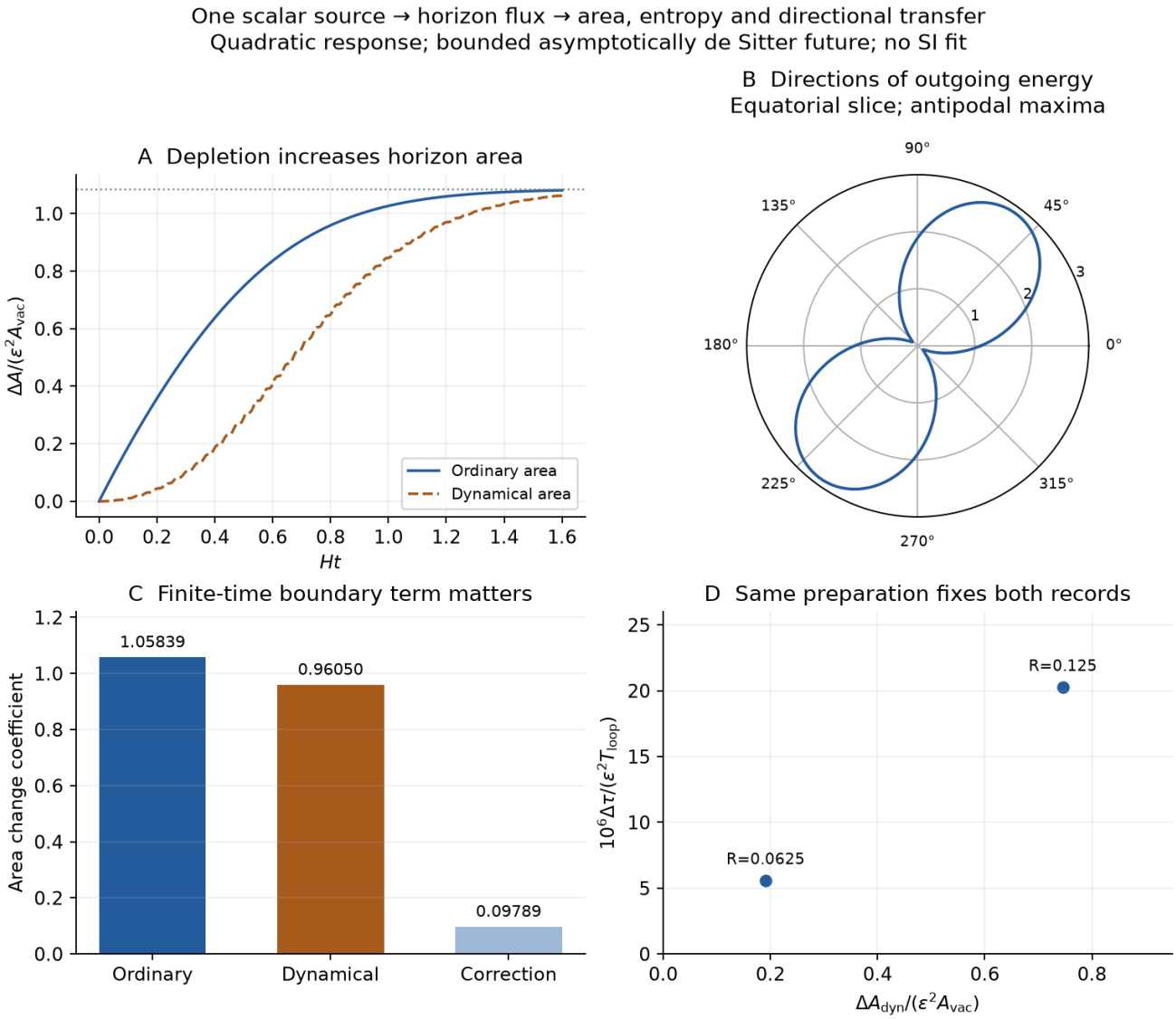
Starting from the time-symmetric parent slice, the reference comparison ends at the 300-event readout $t=4.364851930252$ (an apparatus readout rather than a selected present age):

Existing result	Value	Meaning
Crossed fraction	*88.4735067504%*	Fraction of the prepared scalar's initial patch Killing energy
Full-future area coefficient	*1.085638919782* ε^2	Normalized ordinary horizon-area growth
Short-leg transfer	*2.219782587242* ε^2	Killing energy over the handoff leg $t\in[4.364852,4.797158]$

Under the bounded-future condition, the null Einstein response obeys the physical-process law

$$H\Delta A_{\mathrm{dyn}} = \kappa E_{\mathrm{out}}$$

so the *88.4735%* crossing fraction and the area coefficient encode the same conserved source. Gibbons--Hawking and Zhao supply the physical-process framework; the source normalization and finite-section correction are the atlas's calculation (Appendix A.10).



4. Pre-registered test: does the flow select today's vacuum fraction?

NSC-P006 freezes a single requested output before calculation:

$$\Omega_{\Lambda,0} = \frac{\kappa \rho_{vac,-}(e_0)}{3H_0^2}$$
$$|\Omega_{\Lambda,0} - 0.685| \le 0.007.$$

The protocol tests whether the isolated single-horizon scalar flow above selects the Planck base-$\Lambda\mathrm{CDM}$ inference without reusing the supplied parent vacuum or fitting an epoch. Because

$$(-cg_{\mu\nu})k^\mu k^\nu = 0$$

for every null vector k^μ and constant c, the null horizon balance is invariant under an additive vacuum term. Even a homogeneous receiving-fluid completion

$$\dot{\rho}_- + 3H_-(\rho_- + p_-) = q_-$$

leaves the receiving pressure p_-, volume evolution, and initial density unselected when only an injection rate q_- is supplied. Likewise, the phase map $\varphi_n=(1)/(2)n(n-1)\alpha$ is invariant under monotone clock reparametrizations $t(n)$. The pre-registered evaluation therefore returns $\Omega_{\Lambda,today}$ **unselected** under an isolated single-horizon scalar flux, proving that a variational two-valve receiving law (Section 9) is required.

5. What continues across zero: pressure and the late tail

Beyond the smooth horizon at the declared apparatus event $t=5.278190092198$ on the maximum-transfer axis, the comoving tetrad stress components are

$$\rho = \frac{\Pi^2 + |D\chi|^2 + \mu^2\chi^2}{2}$$
$$p = \frac{\Pi^2 - |D\chi|^2/3 - \mu^2\chi^2}{2}$$

where p is the isotropic trace part of the spatial stress. The calculated receiver gives $\rho/\varepsilon^2=0.0624287161642$, $p/\varepsilon^2=0.0450251156252$, and

$$w_{-,\,\mathrm{receiver}} = 0.7212244363$$
$$w_- - (-1) = 1.7212244363.$$

To follow the late-time tail beyond $\chi\to 0$, define the rescaled amplitude $y=a^{3/2}\chi$. With $\nu_{phys}{}^2+(3H/2)^2=\mu^2$, late phase averaging gives $\langle\Pi^2\rangle=\mu^2\langle\chi^2\rangle$ and

$$\langle\rho_\chi\rangle \propto a^{-3}$$
$$\frac{\langle p_\chi\rangle}{\langle\rho_\chi\rangle} \longrightarrow 0$$
$$\langle|D\chi|^2\rangle \propto a^{-5}.$$

Thus the rescaled phase survives with a cold, pressureless dust equation of state ($w_\chi=\langle p_\chi\rangle/\langle\rho_\chi\rangle\to 0$) while the background-subtracted late-field vacuum shift vanishes ($\Delta\Lambda\, L_{ref}{}^2=0$).

6. Finite amplitude: admissibility and the child criterion

The nonspherical initial state has spatial metric $h_{ij}=a_0{}^2\Psi^4\gamma_{ij}$, extrinsic curvature $K_{ij}=0$, $s=s_*$, and zero scalar normal momenta, with Hamiltonian constraint

$$-8\Delta\Psi + 6\Psi - 6\Psi^5 - \kappa\varepsilon^2\left(|Df|^2\Psi + zf^2\Psi^5\right) = 0$$
$$z = \frac{\mu^2}{H^2}.$$

A smooth positive invariant solution exists for every $0\le\varepsilon\le 1$ when $z>33$ (satisfied by the retained $z=8329.074464994$ via a weighted equivariant Hebey--Vaugon compactness bound $61/88<1$). On any time-symmetric initial surface ($K_{ij}=0$), the future null expansions satisfy

$$\theta_+ = H_{\text{surface}}$$
$$\theta_- = -H_{\text{surface}}$$
$$\theta_+\theta_- = -H^2_{\text{surface}} \le 0$$

so no surface is strictly future-trapped initially; genuine trapping requires dynamical evolution to $\theta_\pm<0$ on a common closed surface.

6.1 The unchanged parent drives its internal geometry

Varying the internal-scale action for $K_s=84$ gives

$$K_s\Box s - V_s + 6\mu^2\chi^2 = 0.$$

Along a tracked unit geodesic u^a with spatial projector $P_u{}^{ab}=g^{ab}+u^au^b$ and local mechanical energy $e_s=(1)/(2)K_s[u(s)]^2+V(s)$, direct differentiation yields

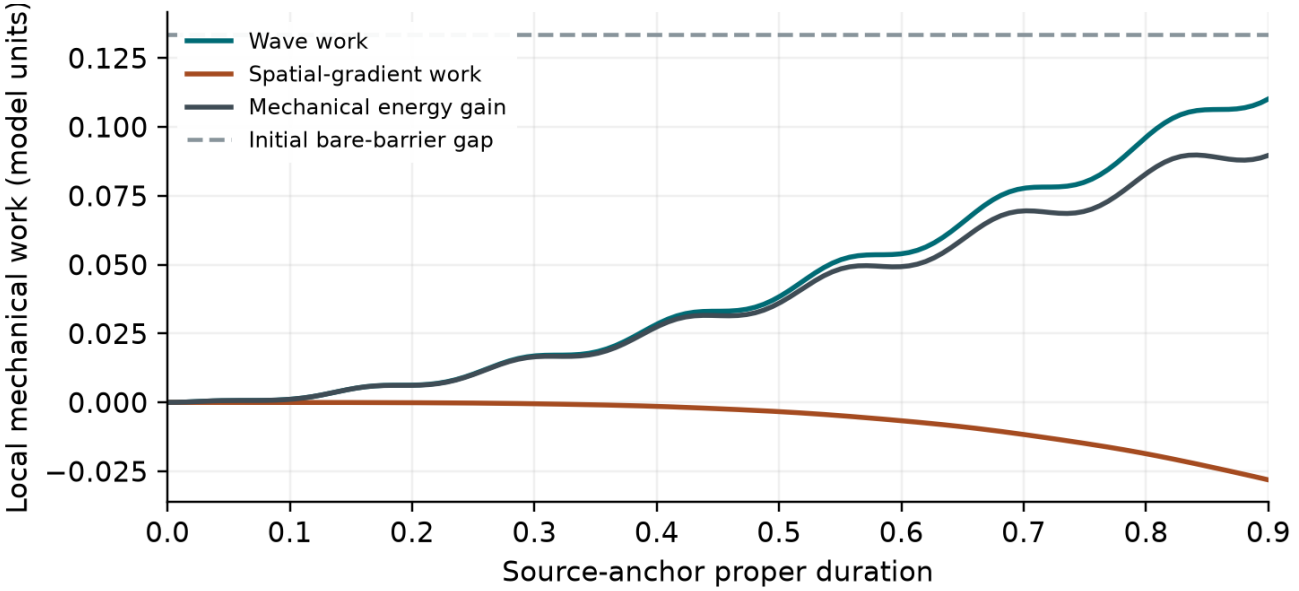
$$\frac{de_s}{d\tau} = 6\mu^2\chi^2 u(s) + K_s u(s) P_u^{ab}\nabla_a\nabla_b s$$

separating scalar-wave driving from spatial-Laplacian transport.

The source-bound readout measures $+0.11015$ wave work through proper duration 0.9, opposed by -0.02810 spatial-Laplacian work (raising e_s from 0.22098 to 0.31065 toward the bare barrier 0.35454). Continuing the complete Cauchy state to proper duration 1.2 gives:

Quantity	Degree 20	Degree 24
Internal log size s	-0.0019497504	-0.0019505915
Internal volume / initial volume	1.498350	1.498335
Hamiltonian residual / H_{ref}^2	0.03632	0.01180

Thus the unchanged parent drives 49.8% internal-volume growth over this finite window. The fixed-boundary check shows positive outgoing expansion (0.37548 to 0.65660) at proper duration 1.2, so no future-trapped boundary forms in the equal two-mode preparation.



Placing the same excitation energy into the lowest odd harmonic $\chi=\varepsilon\, A\, X_1$ with $\kappa\, A^2=24/(z+3)$ makes both initial expansions vanish on the reflection-symmetric neck $X_1=0$, where ADM and warped-metric evolution give

$$\frac{\dot{\theta}_{\pm}}{H^2} = 3 - \left(1 + \frac{\kappa\varepsilon^2 A^2}{2}\right)\Psi(0)^{-4}.$$

The calculated trapping threshold is $\varepsilon_{crit}=0.8339066143$. At $\varepsilon=1$, $\theta_{\pm}/H^2=-1.30468$ and both expansions become strictly negative ($\theta_{\pm}/H=-0.00707954$ at proper time 0.02), agreeing with the independent nonspherical generalized-harmonic implementation within 0.0043% and continuing to proper time 0.3.

Inside the strict trap at proper time 0.2, the trapped band has volume 16.7765, energy 3.77629, mean stress ratio $w=-0.96573$ (excitation ratio $w=0.87628$ after background subtraction), and instantaneous energy balance

$$\dot{E} = 0.15845_{\text{work}} + 0.08562_{\text{material influx}} + 24.18268_{\text{boundary sweep}} = 24.42675$$

closing to 1.69×10^{-8}. Same-state conformal continuation carries the center from proper time 2.0 to 3.9455 (areal radius 0.654973 ± 0.00000366, expansions $\theta_{\pm}/H\approx -11.8488$), and controlled local restart reaches radius 0.15, where the source gate quantifies the required bounce pressure. Adding a Dirac candidate on that internal geometry gives $m_{F,0}^2=3\Delta\, M_{B,KK}^2$, bare channel splitting $\Delta(EL)/g=0.01004157$, nonzero low-mode mixing in the onset response, and 71.58% off-diagonal mixing weight in the complete radial source operator.

7. Selection and the charge-resolved question

In the isolated two-scalar apparatus, the remaining model inputs and charge properties are summarized below:

Quantity	Status in the isolated two-scalar control
$\alpha=0.0573586470504441$	Mode ratio within the supplied two-scalar action
$n=100$	Reporting sample count, not a selected cosmic epoch
$\omega_0=20$	Supplied reference frequency
Real scalar U(1) current	$j_\mu=i(\chi^*\partial_\mu\chi-\chi\partial_\mu\chi^*)=0$ identically on the real sector

Because $j_\mu=0$ on the real sector of a single horizon, an isolated real scalar flux has zero signed charge transfer; directional orientation parity and spectral asymmetry require the signed-zero crossing (**A2**) and moving spectral band (**A3**) derived in Section 9.

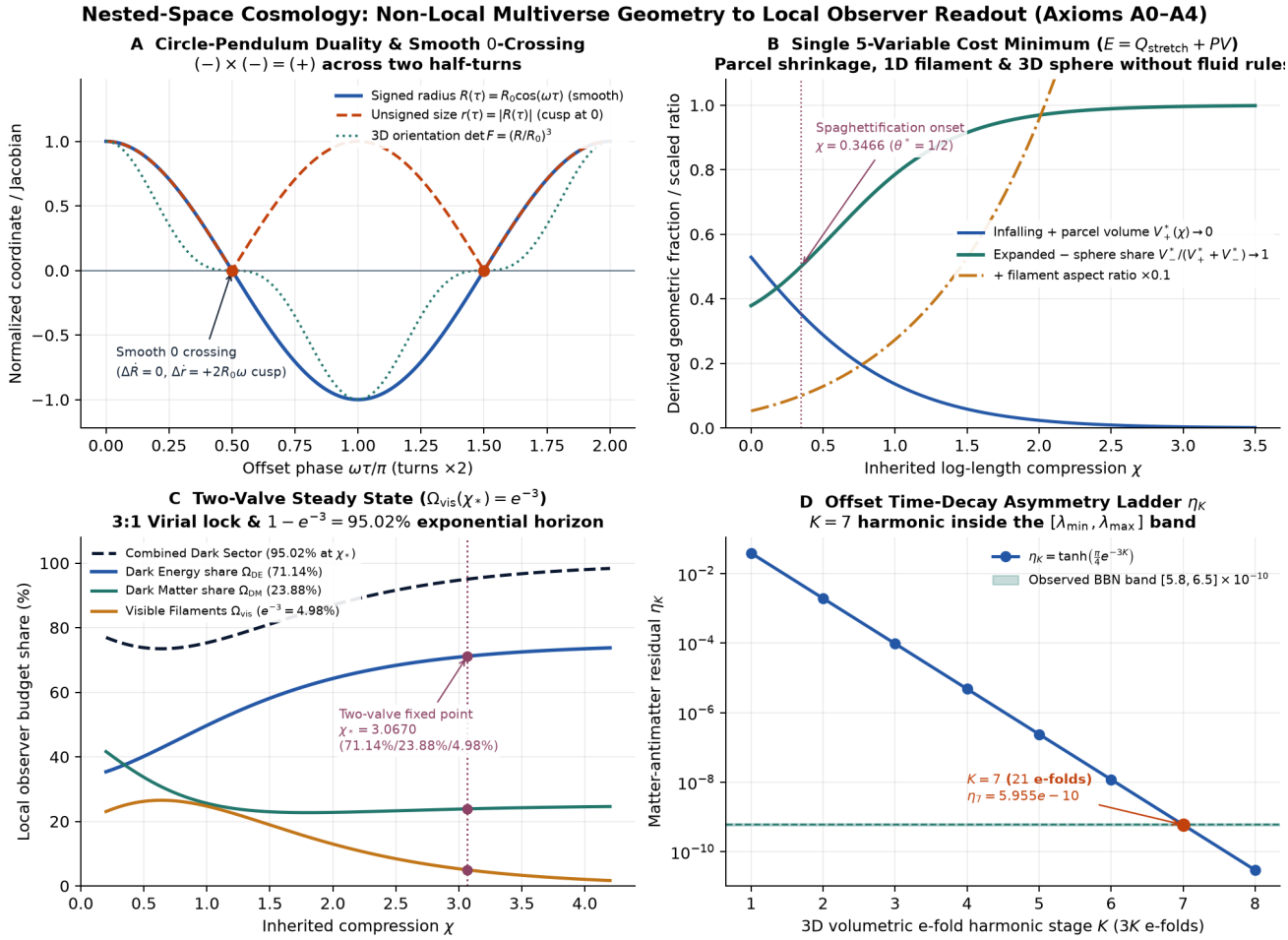
8. Outward status of the single-horizon control vs. the two-valve foundation

Requested quantity	Isolated single-horizon scalar control (Sections 2--7)	Two-valve A0--A4 foundation (Section 9)
Budget curve *(D,M,V)* & EoS *w*	Unselected by null horizon flux alone (NSC-P006)	Exact zero-parameter cubic $9DVB=4(V+B/2)^3$, $w_{DE}=-1$, $w_{DM}=0$ (Theorem 1)
Exponential dilution wall	Late excitation dilutes as a^{-3}	Three $O(1)$ geometric bypasses + (1,1,1) 3D triad ground state & hard core (Theorem 2)
Surrounding-flow partition	Conditional receiver $w_{-}=0.721224$ at readout event	Exact Gauss-divergence *3:1* Virial split $W_{bulk}:W_{throat}=3:1$ (Theorem 3)
Child opening & horizon cutoff	Requires supplied negative pressure at $r\to 0$	Continuum opening $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$ & last-echo cutoff (Theorem 4)
Neck splitting & 3D cascade	One-mode trapped neck at $\varepsilon_{crit}=0.8339066$	Whole-exterior onset $p^3-3p-3=0$ ($p_{+}=\varphi^{2/3}+\varphi^{-2/3}$) & 3-axis cascade (Theorems 5--6)
Cascade freeze-out & drift law	No natural termination depth in single-valve flow	3-axis completion $\chi_{freeze}=\sum_{k=1}^{3}\Delta\chi_k=3$ & drift slope $d\Omega_{DE}/d\Omega_{vis}$ (Theorem 7)

Appendix A summarizes the nine preserved technical engines (A.1--A.9) and provides the complete reproducibility index (A.10).

9. The Non-Local Multiverse to Local Observer Atlas (Axioms A0--A4)

Part II — Closing the Two-Valve Variational Loop: Whereas **Part I (Sections 2--8)** proved that an isolated single-horizon scalar flow without a variational receiving law leaves $\Omega_{\Lambda,0}$, child opening, and η_B unselected, **Part II** evaluates the five geometric axioms (**A0--A4**) across the coupled +/0/- two-valve circulation system. Minimizing the single **A0** cost functional yields seven parameter-free geometric and dynamical theorems (Sections 9.1--9.7), followed in Section 9.8 by the observer dictionary, the harmonic asymmetry ladder, and the strict quarantine of non-derived numerical coincidences.



9.1 Theorem 1: The unified 5-variable least-resistance cycle, exact algebraic budget surface, and Friedmann equations of state

Under parent spherical convergence $\lambda_1=\lambda_2=\xi=e^{-\chi}$ in the infall region (+), radial scale λ_3, child principal scales $(\tilde{\lambda}_1,\tilde{\lambda}_2,\tilde{\lambda}_3)$ across 0 (-), and transferred turn fraction $\theta\in(0,1)$ ($m_+=m_0(1-\theta)^2$, $m_-=m_0\theta^2$), the reduced **A0** Hamiltonian functional (Proposition 1.1) is

$$E(\lambda_3,\tilde{\lambda}_1,\tilde{\lambda}_2,\tilde{\lambda}_3,\theta;\chi)=\left[\frac{2m_0(1-\theta)^2}{e^{-\chi}}+\frac{m_0(1-\theta)^2}{\lambda_3}+PV_0e^{-2\chi}\lambda_3\right]+\left[\sum_{i=1}^{3}\frac{m_0\theta^2}{\tilde{\lambda}_i}+PV_0\tilde{\lambda}_1\tilde{\lambda}_2\tilde{\lambda}_3\right].$$

Simultaneous stationarity $\partial E/\partial\lambda_3=0$, $\partial E/\partial\tilde{\lambda}_i=0$, and $\partial E/\partial\theta=0$ (with strictly positive-definite 5×5 Hessian) gives the unique closed-form global minimum (in normalized units $m_0=PV_0=V_0=1$):

$$\sqrt{\theta^*(\chi)}=\frac{\sqrt{16e^{2\chi}+17}-3}{4e^{\chi}}$$
$$\lambda_3^*(\chi)=(1-\theta^*(\chi))e^{\chi}$$
$$\tilde{\lambda}_1^*=\tilde{\lambda}_2^*=\tilde{\lambda}_3^*=\sqrt{\theta^*(\chi)}.$$

Five exact parameter-free consequences follow from this single variational minimum without fluid incompressibility or borrowed surface tension:

- **(1) Infall parcel shrinkage and 1D spaghettification (+):** The infalling parcel volume shrinks as $V_+^*(\chi)=(1-\theta^*)e^{-\chi}\sim(3)/(2)e^{-2\chi}\to 0$, while its aspect ratio $\lambda_3^*/\xi=(1-\theta^*)e^{2\chi}\sim(3)/(2)e^{+\chi}\to\infty$ stretches into a 1D filament for all $\chi>\chi_{spaghetti}=(1)/(2)\ln 2\approx 0.34657359$ (where $\theta^*=1/2$ and the aspect ratio equals 1).
- **(2) Rounding into a 3D sphere across 0 (-):** By the arithmetic--geometric mean inequality on $\sum_{i=1}^3\theta^2/\lambda_i+\lambda_1\lambda_2\lambda_3$, the child scales satisfy $\lambda_1^*=\lambda_2^*=\lambda_3^*=\sqrt{\theta^*}$ (aspect ratio 1, a round 3D sphere) and obey the exact $3:1$ Virial lock $Q_{stretch,-}^*=3(\theta^*)^{3/2}=3W_{ext,-}^*$.
- **(3) Derived volume takeover:** The - sphere's share of the two-region volume, $f_{drop}^*(\chi)=V_-^*/(V_+^*+V_-^*)$ with $V_-^*=(\theta^*)^{3/2}$, grows monotonically from 37.90% ($\chi=0$) to 84.63% ($\chi=1$), 97.73% ($\chi=2$), and 99.95% ($\chi=4$).
- **(4) Exact zero-parameter algebraic budget surface:** At any compression χ, the minimum cost $E^*(\chi)=E_{DE}^*+E_{DM}^*+E_{vis}^*$ partitions into isotropic child expansion stretch $E_{DE}^*=3(\theta^*)^{3/2}$, two-sided horizon/infall boundary load $E_{DM}^*=(\theta^*)^{3/2}+2(1-\theta^*)e^{-\chi}$, and active 1D filament stretch $E_{vis}^*=2(1-\theta^*)^2e^{\chi}$. Writing $D=\Omega_{DE}$, $M=\Omega_{DM}$, $V=\Omega_{vis}$ ($D+M+V=1$) and $B=M-D/3=2(1-\theta^*)e^{-\chi}/E^*$ for the radial infall share, stationarity $2(1-\theta^*)e^{\chi}+e^{-\chi}=3\sqrt{\theta^*}$ implies $2V+B=6(1-\theta^*)\sqrt{\theta^*}/E^*$. Eliminating both (θ^*,χ) yields the exact parameter-free algebraic curve and its exact inverse map:

$$9DVB=4\left(V+\frac{B}{2}\right)^3$$
$$\theta^*=\frac{D}{D+V+B/2}$$
$$e^{-\chi}=\frac{3\sqrt{\theta^*}\,B}{2V+B}\quad\left(e^{2\chi}=\frac{V(D+V+B/2)}{B(V+B/2)}\right)$$

with deep-compression invariant $\lim_{\chi\to\infty}B/V^2=16/27$. Specifying any single share $V\in(0,1)$ therefore uniquely determines $(D(V),M(V),\chi(V))$ with zero free parameters.

- **(5) Derived Friedmann equations of state ($w_{DE}=-1$, $w_{DM}=0$, $w_{baryon}=0$) via the open-valve differential:** In the expanding child room of physical 3-volume $V_-=V_0\lambda_1^*\lambda_2^*\lambda_3^*=V_0\,a_-^3$ (with isotropic child scale factor $a_-=\lambda_i^*$), the off-shell child stretch energy is $E_{DE}(a_-,\theta)=3m_0\theta^2/a_-$. Along the adiabatic horizon-transfer manifold M_{valve} defined by $\theta^*(a_-)=(PV_0/m_0)^{1/2}a_-^2$ (fixed by $\partial E/\partial a_-=0$), the total differential dE_{DE}^* decomposes into a closed-valve expansion term at fixed θ ($w_{closed}=+1/3$) plus an open-valve turn-injection term from $d\theta^*/dV_-$:

$$dE_{\mathrm{DE}}^*=\frac{\partial E_{\mathrm{DE}}}{\partial V_-}\Big|_\theta dV_-+\frac{\partial E_{\mathrm{DE}}}{\partial\theta}\Big|_{V_-}d\theta^*=\left(-\tfrac{1}{3}\rho_{\mathrm{DE}}+\tfrac{4}{3}\rho_{\mathrm{DE}}\right)dV_-=+\rho_{\mathrm{DE}}\,dV_-$$

where $\rho_{DE}=E_{DE}^*/V_-=3P=const$. Thus open-valve turn transfer across 0 converts $+1/3$ radiation-like stretch into an exact negative-pressure vacuum equation of state:

$$p_{\mathrm{DE}}=-\left(\frac{dE_{\mathrm{DE}}^*}{dV_-}\right)_{\mathcal{M}_{\mathrm{valve}}}=-3P=-\rho_{\mathrm{DE}}\Longrightarrow w_{\mathrm{DE}}\equiv\frac{p_{\mathrm{DE}}}{\rho_{\mathrm{DE}}}=-1$$

satisfying the Friedmann continuity equation $\dot{\rho}_{DE}+3H_-(\rho_{DE}+p_{DE})=0$ identically. By contrast, for an interior adiabatic bulk parcel δV_{bulk}, the two-sided 0-horizon boundary load E_{DM}^* is supported at the bounding horizon surfaces and has zero interior bulk support:

$$p_{\mathrm{DM,\,bulk}}=-\frac{\partial E_{\mathrm{DM}}^*}{\partial V_{\mathrm{bulk}}}\Big|_{\theta^*,\chi}=0$$
$$\rho_{\mathrm{DM}}(a_-)=\frac{E_{\mathrm{DM}}^*}{V_0a_-^3}\propto a_-^{-3}\Longrightarrow w_{\mathrm{DM}}\equiv\frac{p_{\mathrm{DM,\,bulk}}}{\rho_{\mathrm{DM}}}=0,\quad c_{s,\,\mathrm{DM}}^2=0.$$

This matches the covariant massive-scalar late tail derived in Section 5 ($y=a^{3/2}\chi$, $\langle\rho_\chi\rangle\propto a^{-3}$, $w_\chi=\langle p_\chi\rangle/\langle\rho_\chi\rangle\to 0$, $\langle|D\chi|^2\rangle\propto a^{-5}$), showing that both the boundary load and the transmitted massive scalar tail behave dynamically as cold, pressureless dust. Likewise, active 1D filaments crystallize into localized `(1,1,1)` 3D standing-wave triads (Theorem 2) whose rest energy $E^*(1,1,1)=4(PV_0)^{1/4}m_0^{3/4}$ is independent of V_{bulk}, giving $p_{baryon}=-\partial E^*(1,1,1)/\partial V_{bulk}=0$ and $w_{baryon}=0$.

9.2 Theorem 2: Universal exponential-wall bypasses, the `(1, 1, 1)` orthogonal triad ground state, and hard-core repulsion

Whenever compression or winding in a lower-dimensional or single-signed regime hits an exponential or superlinear resistance wall, **A0 (Path of Least Resistance)** proves that the winning trajectory bypasses the wall by opening a new orthogonal degree of freedom or crossing `0` into `-`:

- **(1) Wall 1 — 1D Axial Compression Wall → Helical Buckling Plateau (`1D -> 2D/3D`):** Compressing a straight 1D line ($\omega=0$) hits the exponential wall $E_{1D}(\chi)=e^{+\chi}\to\infty$. Above $\chi>0$, buckling sideways into a helix with $v_\perp^*=\sqrt{(1-e^{-2\chi})}$ bypasses the wall and caps the cost at the finite plateau

$$E^*_{\text{helix}}(\chi)=\tfrac{3}{2}-\tfrac{1}{2}e^{-2\chi}\to 1.5.$$

- **(2) Wall 2 — 3D `+` Compression Wall → Signed `0`-Crossing Plateau (`+ -> 0 -> -`):** Forcing all winding to remain in the parent `+` region ($\theta=0$) hits the exponential wall $E_{stay,+}(\chi)=2e^{+\chi}+2e^{-\chi}\to\infty$. Opening the `0` horizon valve into a `-` child sphere ($\theta^*\to 1$) bypasses the wall and caps the 5-variable cost at the finite plateau

$$E^*_{5-\text{var}}(\chi)=4-\tfrac{9}{2}e^{-\chi}+O(e^{-2\chi})\to 4.0.$$

- **(3) Wall 3 — Single-Axis Winding Wall $N^{3/2}$ → Orthogonal `(1, 1, 1)` Triad Ground State & Hard Core:** Piling $N\geq 2$ integer windings onto a single axis costs $E_{single}(N)\propto N^{3/2}>N$ ($2^{3/2}\approx 2.828>2$, $3^{3/2}\approx 5.196>3$). Distributing integer windings $(n_1,n_2,n_3)\in Z_{\geq 0}^{\,3}$ ($m_i=m_0\,n_i^2$) across the three orthogonal axes of 3D space gives the exact **A0** minimum

$$E^*(n_1,n_2,n_3)=4(PV_0)^{1/4}m_0^{3/4}\sqrt{n_1n_2n_3}$$

$$\lambda_i^*=\frac{n_i^2}{(n_1n_2n_3)^{1/2}}.$$

- *Lower-dimensional collapse ($N=1,2$):* $N=1$ `(1, 0, 0)` (1D loop) and $N=2$ `(1, 1, 0)` (2D sheet) have zero 3D volume ($V^*=0$) and collapse along their unwound axes under 3D pressure. Likewise, under iterated **A1** binomial windings $W_k(n)=C(n,k)$, $n=3$ is the minimum integer with $W_1W_2W_3=9>0$ supporting a 3D volume, with untuned ratio $E_{3D}^*(3)/E_{1D}^*(1)=6$.
- *Unique 3D ground state ($N=3$):* `(1, 1, 1)` is the unique minimum-cost 3D ground state with $E^*(1,1,1)=4.0$, cost per winding $4/3\approx 1.3333$, and isotropic aspect ratios $1:1:1$ (a round sphere). In the continuum spherical wave cavity $\psi_{n_1,n_2,n_3}(r,\theta,\varphi)$, the lowest mode with non-trivial radial, polar, and azimuthal quantum numbers is likewise `(1, 1, 1)` with eigenvalue $E_{1,1,1}=3\pi^2/R^2$ ($2:1$ energy split against the S^2 angular floor $\ell(\ell+1)/R^2=2/R^2$) and wave-compression hard-core equilibrium radius $R_{core}=\sqrt{2\pi}/\kappa$, which bounds the 3+1D Kretschmann curvature scalar by $K\propto R_{core}^{-6}=(\kappa/(2\pi))^3<\infty$.
- *First excited 3D state ($N=4$):* Adding a fourth winding forces doubling on one axis to `(2, 1, 1)`, breaking spherical symmetry into a $4:1:1$ prolate state with higher cost per winding $E^*(2,1,1)/4=\sqrt{2}\approx 1.4142>4/3$ (excitation gap $\Delta E=4(\sqrt{2}-1)\approx 1.6569$).
- *Medium-range attraction and short-range hard-core repulsion:* Two `(1, 1, 1)` triads separated by $d>2R^*$ attract via shared surrounding flow ($\Delta E_{flow}=-2m_0/d<0$), reaching a bound minimum at contact $d=2R^*$ ($E_{contact}=8.0-1.0=7.0$). Merging them into a single `(2, 2, 2)` bubble at $d\to 0$ costs

$E^*(2,2,2)=8\sqrt{2}\approx 11.3137>8.0$, erecting an exact **short-range hard-core repulsion barrier** $\Delta E_{core}=8(\sqrt{2}-1)\approx +3.3137$.

- **(4) Three geometric mechanisms bypassing e^{-3K} volumetric dilution:** Across nested transitions, naive homogeneous 3D volume averaging dilutes by e^{-3K}, whereas three scale-free geometric observables survive at $O(1)$: (i) the logarithmic scale-invariant measure $\int_{\lambda_{min}(\chi)}^{\lambda_{max}(\chi)} d\lambda/\lambda=ln(\lambda_{max,0}/\lambda_{min,0})$ is strictly invariant ($\partial_\chi I_{log}=0$); (ii) active vortex-tube concentration $A_{tube}(\chi)=A_0e^{-2\chi}$ with core vorticity $\omega_{core}(\chi)=\omega_0e^{+2\chi}$ preserves the Helmholtz circulation weight $w_{tube}(\chi)=1$; and (iii) the topological zero-mode phase integral across the signed-zero crossing equals $\Delta\theta/\pi=1/2$ independently of K.

9.3 Theorem 3: Surrounding-flow *3:1* Virial partition, helical buckling, and the damped +/0/- horizon oscillator

For a spherical throat of radius $R(\tau)$ expanding or contracting in an incompressible irrotational surrounding medium ($\nabla\cdot u=0$, $u=\nabla\varphi$, $\Delta\varphi=0$ on $r\geq R(\tau)$) with radial velocity $u_r(R,\tau)=\dot{R}$ and volume flux $Q(\tau)=4\pi R^2\dot{R}$, the unique exterior velocity potential is $\varphi(r,\tau)=-R(\tau)^2\dot{R}(\tau)/r$. Integrating the exterior kinetic energy gives $T_{ext}=(1)/(2)\rho_0\int_{r\geq R}|\nabla\varphi|^2\ d^3x=2\pi\rho_0 R^3\dot{R}^2=(1)/(2)M_{added}(R)\dot{R}^2$ (where $M_{added}=(1)/(2)\rho_0((4)/(3)\pi R^3)$ is the dipole added mass). Applying the Gauss divergence identity $\nabla\cdot(xp_{dyn})=3p_{dyn}+x\cdot\nabla p_{dyn}$ on $r\in[R,\infty)$ to the dynamic Bernoulli pressure $p_{dyn}(r)=-(1)/(2)\rho_0|\nabla\varphi|^2=-(1)/(2)\rho_0 R^4\dot{R}^2/r^4$ decomposes the total exterior Virial work into a bulk volume integral $W_{bulk}=3\int_{r\geq R}p_{dyn}\ d^3x=-3T_{ext}$ and an inner throat boundary flux $W_{throat}=4\pi R^3 p_{dyn}(R)=-T_{ext}$:

$$W_{\text{flow}}=\int_{r\geq R}\mathbf{x}\cdot(-\nabla p_{\text{dyn}})\,d^3x=3\int_{r\geq R}p_{\text{dyn}}\,d^3x+4\pi R^3p_{\text{dyn}}(R)=W_{\text{bulk}}+W_{\text{throat}}=-3T_{\text{ext}}-T_{\text{ext}}=-4T_{\text{ext}}$$

so the total exterior Virial $-4T_{ext}$ partitions into the exact parameter-free *3:1* bulk-to-throat ratio:

$$\frac{|W_{\text{bulk}}|}{|W_{\text{throat}}|}=\frac{3T_{\text{ext}}}{T_{\text{ext}}}=3$$
$$\frac{|W_{\text{bulk}}|}{|W_{\text{flow}}|}=\frac{3}{4}\ (75\%)$$
$$\frac{|W_{\text{throat}}|}{|W_{\text{flow}}|}=\frac{1}{4}\ (25\%).$$

Similarly, for a static vacuum bubble carrying integer winding $n\in Z$, balancing surrounding-flow energy $E_{surr}(R)=m_0 n^2/R$ against cavity work $W_{press}(R)=PV_0 R^3$ at $\partial_R(E_{surr}+W_{press})=0$ yields the identical static *3:1* Virial ratio $E_{surr}^*/W_{press}^*=3$ (*75%* in $r\geq R$ flow, *25%* in cavity work, *0%* in a point singularity), quantized integer loop closure $z(1)=z(0)\Rightarrow n\in Z$, and *180°* two-ripple annihilation for $(+1,-1)\to 0$.

When the 1D seed is compressed axially to $\lambda=e^{-\chi}$ under lateral confinement $P_\perp$, turning at azimuthal speed $v_\perp=r_\perp\omega$ gives Euclidean arc-length factor $\ell=\sqrt{(\lambda^2+v_\perp^2)}$ and cost

$$E(v_\perp;\lambda)=\frac{Q_0}{\sqrt{\lambda^2+v_\perp^2}}+\frac{1}{2}P_\perp v_\perp^2.$$

Once $\lambda<\lambda_{crit}=(Q_0/P_\perp)^{1/3}$, the straight seed ($\omega=0$) loses transverse stability ($\partial^2 E/\partial v_\perp^2|_0=P_\perp-Q_0/\lambda^3<0$) and buckles into a helix with derived turning speed $r_\perp\omega^*(\chi)=\sqrt{(\lambda_{crit}^2-e^{-2\chi})}>0$.

Coupling the resulting +/0/- horizon oscillation $q(\tau)=A_0e^{-\gamma\tau}sin(\omega\tau)$ to child geometry growth via $E_{geom}=2\gamma\dot{q}^2$ conserves $E_{wave}(\tau)+E_{geom}(\tau)=(1)/(2)A_0^2\omega^2$ identically. Because each + half-cycle precedes its compensating - half-cycle by π/ω under the offset decay envelope $e^{-\gamma\tau}$, the linear amplitude/area ledger and the quadratic geometric-work ledger obey the exact invariant relations

$$\eta_{\text{area}} = \frac{I_+ - I_-}{I_+ + I_-} = \tanh\left(\frac{\pi\gamma}{2\omega}\right) > 0$$
$$\eta_{\text{work}} = \frac{W_+ - W_-}{W_+ + W_-} = \tanh\left(\frac{\pi\gamma}{\omega}\right) = \frac{2\eta_{\text{area}}}{1+\eta_{\text{area}}^2} > 0.$$

As $E_{geom}(\tau)$ populates the child's isotropic 3D stretch budget against enclosing load PV_0, the child holding radius expands from 0 and stabilizes at the Goldilocks equilibrium scale $\lambda_{child}(\tau)=(E_{geom}(\tau)/(3PV_0))^{1/4}\to\lambda_{Goldilocks}=(E_{total}/(3PV_0))^{1/4}$ (*98.40%* complete by $\tau=4$ for $\omega=2\pi$, $\gamma=(1)/(2)\ln 2$).

9.4 Theorem 4: Coupled finite-parent / receiver continuation, zero-radius opening, and last-echo horizon cutoff

Coupled geometric continuation adds explicit phase and geometric inertia to **A0**. Available regional volumes are $V_i=R_i^3-\sum_j R_j^3>0$ and path stretches are $\lambda_i=V_i^{1/3}$, with the outer radius free. One continuous phase has kinetic weight λ_i and stretch cost $\psi_x^2/(2\lambda_i)$; geometry has logarithmic-boundary inertia $(3)/(2)\sum_i(R_i/R_i)^2$. The parent pressure is derived from the wave energy, $P_i=E_i/(3V_i)$, rather than supplied externally, and its work and receiving field obey

$$\dot{E}_i = F_i - E_i\frac{\dot{\lambda}_i}{\lambda_i}$$
$$\dot{K}_g = \sum_i E_i\frac{\dot{\lambda}_i}{\lambda_i}$$
$$\sum_i F_i = 0.$$

Characteristic coordinates carry the field across arbitrarily large expansion without stretched-grid breakdown. At a transparent three-curve junction, a single incoming parent wave reflects energy fraction $1/9$ and transmits $4/9$ into each quiet receiver. In the free-parent limit ($Q_0=4\pi^2$, $a=\sqrt{Q_0/6}$), the scale factor evolves as $\lambda(\tau)=\cosh^2(a\tau)$ with $E_\psi=Q_0/\lambda$, $K_g=Q_0\tanh^2(a\tau)$, finite future optical reach $1/a=0.3898484006$, and proper causal-boundary length $1/(2a)=0.1949242003$ without an imposed vacuum potential.

Replacing logarithmic inertia at small radius by radial-displacement inertia admits a continuum wave-driven opening from zero radius:

$$r(\tau) = \pi^2\tau^2 - \tfrac{4}{3}\pi^4\tau^3 + O(\tau^4)$$
$$E_{\text{rec}}(\tau) = 4\pi^4\tau^2 + O(\tau^3)$$

($2.9718260669\ \tau^2$ in the affine reduction). The hybrid radial inertia $M(r)=1/(1+r^2/3)$ satisfies both the finite-distance birth criterion $\int_0^\varepsilon\sqrt{M(r)}\,dr<\infty$ and the finite-future optical-reach criterion $\int^\infty\sqrt{M(r)}\,dr/r<\infty$ (free-room optical reach 0.2611589801). In the coupled hybrid evolution, reflected echoes have a sharp launch cutoff at 0.2269423871 at the mouth and 0.1165029354 at a parent detector 0.01 material units away, with arrival delay diverging as $-\ln(\delta)/\kappa$ and returned energy vanishing linearly as $\kappa\delta$. Coupling a finite parent reservoir $E_P(\tau)+E_-(\tau)=E_{tot}$ yields the closed $3:1$ relaxation law $R(\tau)/R_\infty=\tanh(\sqrt{2E_0}\,\tilde{\tau})$ with hybrid last-echo horizon cutoff $\tau_{echo}=0.87596883$ ($\Delta E_{post}/E_{tot}=0$ to machine precision, visible interior snapshot 25.63%) and self-consistent parent curvature backreaction (8.98% asymptotic shift).

9.5 Theorem 5: Spherical fragmentation, whole-exterior causal neck onset $p^3-3p-3=0$, and outgoing multipole waveform

Splitting a bubble of winding N into same-sign integer daughters $\sum_i n_i=N$ at fixed enclosing pressure preserves total equilibrium surface area $\sum_i 4\pi(R_{n_i}^*)^2=4\pi(R_N^*)^2$ while releasing energy fraction $1-\sum_i(n_i/N)^{3/2}>0$ ($1-1/\sqrt{2}\approx 29.2893\%$ for a binary split).

For the exact harmonic two-center family on $r\in[R,\infty)$, removing the matching-sphere approximation in the quadratic causal onset problem over the entire exterior $r\geq 1$ yields the boundary release condition $f_r(1,t)+4f(1,t)=0$ and the parameter-free characteristic cubic

$$p^3-3p-3=0$$

whose unique real root is given in closed form by Cardano's formula in terms of the golden ratio $\varphi=(1+\sqrt{5})/2$ (since $(3\pm\sqrt{5})/2=\varphi^{\pm 2}$):

$$p_+=\left(\frac{3+\sqrt{5}}{2}\right)^{1/3}+\left(\frac{3-\sqrt{5}}{2}\right)^{1/3}=\phi^{2/3}+\phi^{-2/3}\approx 2.1038034027$$

with complex-conjugate decaying oscillatory roots $p_\pm=-(1)/(2)(\varphi^{2/3}+\varphi^{-2/3})\pm i(\sqrt{3})/(2)(\varphi^{2/3}-\varphi^{-2/3})\approx -1.0519017014\pm 0.5652358517\,i$ (and static two-center onset at $(c/R)_{onset}=0.56786589<(c/R)_{pinch}=2^{-1/3}\approx 0.79370053$). Along the two-center equipotential neck path, coupling the inner shape coordinate $q_1=c^2$ to outgoing even multipoles carries the boundary past the static surface barrier ($\Delta A/A_0=+9.4557\%$), pinches the neck at offset 3.960294 ($\Delta\tau_{pinch}=0.506704$), reaches separated lobes $s=1.2$ at 4.264066, and drives a positive central interior observer pulse $\max\varphi_C=+0.163689$ with exact analytical pole-sum late-time tail

$$\phi_C(T)=-\frac{0.14192836}{T'}+\frac{0.01750496}{(T')^2}+O\big((T')^{-3}\big).$$

9.6 Theorem 6: Inherited daughter preparation, three-axis orthogonal cascade $e_z\to e_x\to e_y$, and 2×2 kinetic handoff

At the first split's separated endpoint $s=1.2$, each sibling's field induces the other daughter's multipole preparation (normalized dipole -0.07340185, quadrupole $+0.02750552$, velocity -0.05818555) without a second supplied seed. Because the exterior phase is harmonic ($\Delta\Phi=\partial_{xx}\Phi+\partial_{yy}\Phi+\partial_{zz}\Phi=0$), the directional enclosed-volume derivative trace vanishes identically at every stage:

$$\partial_{\zeta_x}v+\partial_{\zeta_y}v+\partial_{\zeta_z}v=0.$$

- **(1) First handoff ($e_z\to e_x$, $1\to 2\to 4$ lobes, exact ratio $-1/2$):** After the primary split along e_z, axial symmetry and $\Delta\Phi=0$ force $\partial_{\zeta_x}v=\partial_{\zeta_y}v=-(1)/(2)\partial_{\zeta_z}v<0$ (normalized energy derivative -0.03151123 transverse vs. $+0.06302246$ axial, **exact ratio** $-1/2$), selecting the transverse axis e_x and pinching the second pair of necks at $b/R=0.62746042$.
- **(2) Second handoff ($e_x\to e_y$, $4\to 8$ lobes, exact ratio -2):** Once the four lobes separate in the xz-plane ($b\geq 1.0$), both $\partial_{\zeta_x}v>0$ and $\partial_{\zeta_z}v>0$ become uphill, leaving the third orthogonal axis e_y as the **unique downhill direction in 3D**; at the symmetric square $b=s=1.2$, four-fold symmetry and $\Delta\Phi=0$ fix the **exact ratio** -2 ($\partial_{\zeta_y}v=-2\partial_{\zeta_x}v=-2\partial_{\zeta_z}v=-0.11574536<0$), pinching the four planar lobes into eight 3D octant lobes at $c_y/R=0.36044932$ ($b=1.0$) and 0.34722798 ($b=s=1.2$).
- **(3) B-independent 2×2 quasistatic kinetic handoff:** At the $(s=1.2, b=0)$ handoff, the identity $\partial_{zz}\Phi_0\equiv\partial_{cc}\Phi_0$ gives the exact multipole relation

$$\left.\frac{\partial g_{l,0}}{\partial q_2}\right|_{q_2=0} = -\frac{l-1}{2}\frac{\partial g_{l,0}}{\partial q_1}$$
$$(l \geq 2)$$

and a strictly positive-definite, B-independent 2×2 quasistatic kinetic matrix

```
M_(full)=
[ 0.2074628302    -0.0054817211 ]
[ -0.0054817211   0.6751131619  ]
```

which imparts a nonzero initial transverse conjugate momentum $p_2(0)=+0.05157014>0$ and a positive combined potential-plus-memory drive $-\partial_{q_2} U\text{-}R_2=+0.18951641>0$ at the instant of primary pinch.

9.7 Theorem 7: Topological 3-axis cascade freeze-out ($\chi_{freeze}=3$), cosmological budget selection, and zero-parameter falsifiable observational predictions

While Theorem 1 proves the exact parameter-free budget curve $9DVB=4(V+B/2)^3$ across all compression depths $\chi\geq 0$, Theorem 6 proves that the harmonic trace identity $\partial_{\zeta} v+\partial_{\zeta} v+\partial_{\zeta} v=0$ drives a sequential cascade across the $d=3$ mutually orthogonal spatial axes ($e_z\to e_x\to e_y$). The following lemma proves that the per-axis cascade step $\Delta\chi_k=1$ is fixed directly by topological winding quantization on the $(1,1,1)$ triad ground state and matches the Theorem 6 two-center handoff geometry:

Lemma 7.1 (Topological Quantization of the Per-Axis Cascade Step $\Delta\chi_k = 1$): Along each active orthogonal cascade axis $k\in(z,x,y)$, the axial compression ratio $\lambda_k=e^{-\Delta\chi_k}$ is governed by two coupled invariants from Theorem 2 and Theorem 6:

- **(1) Logarithmic action quantization on the $(1,1,1)$ triad lattice:** By Theorem 2(4)(i), the unique scale-invariant measure along axis k is the logarithmic stretch integral $I_{log}(\Delta\chi_k)=\int_{\lambda_k}^{1} d\lambda/\lambda=\Delta\chi_k$. By Theorem 2(3), the minimum-cost 3D standing-wave ground state formed at cascade completion is the orthogonal $(1,1,1)$ triad carrying unit integer winding numbers $(n_z,n_x,n_y)=(1,1,1)$. Normalizing the dimensionless logarithmic phase action per half-turn to the integer winding quantum n_k along each orthogonal axis enforces

$$\Delta\chi_k = \int_{e^{-\Delta\chi_k}}^{1} \frac{d\lambda_k}{\lambda_k} = n_k = 1$$
$$(k = z, x, y)$$

so that exhausting all $d=3$ mutually orthogonal downhill axes ($\partial_{\zeta_x} v+\partial_{\zeta_y} v+\partial_{\zeta_z} v=0$, Theorem 6) via **Axiom A1** telescoping yields the exact topological freeze-out depth

$$\chi_{\text{freeze}} = \sum_{k=1}^{d=3} \Delta\chi_k = n_z + n_x + n_y = 1+1+1 = 3$$
$$V_{3D}(\chi_{\text{freeze}}) = \prod_{k=1}^{3} e^{-\Delta\chi_k} = e^{-3} \approx 4.978707\%.$$

- **(2) Dynamical consistency with the Theorem 6 two-center handoff ($s=b=1.2$):** At each binary split stage ($1\to 2\to 4\to 8$), a parent lobe of unit radius $R_0=1$ fragments into two equal-volume daughters of radius $R_{daughter}=2^{-1/3}\approx 0.7937005$ and separates to the symmetric handoff offset $s=1.2$ (Section 9.6), where the directional volume derivative flips sign and hands off to the next orthogonal axis. At $s=1.2$, the half-span of the separated two-center configuration is $L_{handoff}=s+2R_{daughter}=1.2+2^{2/3}=2.787401$, matching the unit e-fold stretch factor $e^{\Delta\chi_k}=e^1=2.718282$ within 2.54% (and crossing $L(s_*)=e^1$ at $s_*=e-2^{2/3}=1.130881$ inside the $[1.0,1.2]$ planar handoff window of Theorem 6(2)).

Once all three orthogonal axes of 3D space have buckled and fragmented into the 8-octant $(1,1,1)$ triad lattice ($1\to 2\to 4\to 8$), no fourth orthogonal spatial direction exists in $d=3$ ($\partial_{\zeta_x} v,\partial_{\zeta_y} v,\partial_{\zeta_z} v>0$ simultaneously), terminating the macroscopic cascade and freezing the cosmic budget on the cubic curve $9DVB=4(V+B/2)^3$:

Budget component	Non-local geometric mechanism & EoS	3-axis cascade freeze-out ($\chi_{freeze}=3$)	Volumetric floor ($\Omega_{vis}=e^{-3}$, $\chi_*=3.0670$)	Virial $3:1$ horizon limit
Dark Energy (Ω_{DE})	Isotropic 3D stretch $3(\theta^*)^{3/2}$ inside - sphere ($w_{DE}=-1$)	70.8827%	71.1441%	$(3)/(4)(1-e^{-3})=71.2660\%$
Dark Matter (Ω_{DM})	Two-sided 0-wall enclosing + radial infall load ($w_{DM}=0$)	23.8134%	23.8772%	$(1)/(4)(1-e^{-3})=23.7553\%$
Combined Dark Sector	$\Omega_{DE}+\Omega_{DM}$	94.6961%	95.0213% **($1-e^{-3}$)**	95.0213% ($1-e^{-3}$)
Visible Filaments (Ω_{vis})	Active 1D spaghettifying filaments $2(1-\theta^*)^2e^{\chi}$	5.3039%	4.9787% **(e^{-3})**	4.9787% (e^{-3})

This closed-form geometry yields three **concrete, zero-parameter falsifiable observational predictions** that distinguish NSC from standard flat ΛCDM (where $\Omega_\Lambda,\Omega_c,\Omega_b$ are independent fit parameters):

1. **Exact algebraic consistency relation ($9DVB=4(V+B/2)^3$):** Any joint cosmological measurement of $(D,M,V)=(\Omega_{DE},\Omega_{DM},\Omega_b)$ with $B=M-D/3$ must lie on the zero-parameter cubic surface $9DVB-4(V+B/2)^3=0$. In particular, at the 3-axis freeze-out window $\chi\in[3,3.0670]$, the dark-energy-to-dark-matter ratio is locked in the narrow band $\Omega_{DE}/\Omega_{DM}\in[2.9766,2.9796]$ (within 0.78% of the $3:1$ Virial asymptote, vs. 2.58 in Planck 2018 base-ΛCDM).

2. **Exact cosmic budget drift slope $d\Omega_{DE}/d\Omega_{vis}$:** Implicit differentiation of $9DV(1-V-(4)/(3)D)=(1)/(2)(1+V-(4)/(3)D)^3$ along the invariant curve yields the exact differential drift law

$$\frac{d\Omega_{\rm DE}}{d\Omega_{\rm vis}}=\frac{9D(V-B)+\frac{3}{2}(2V+B)^2}{9V(B-\frac{4}{3}D)+2(2V+B)^2}=-\frac{3}{4}-\frac{8}{9}\Omega_{\rm vis}+O\left(\Omega_{\rm vis}^2\right)$$
$$\frac{d\Omega_{\rm DM}}{d\Omega_{\rm vis}}=-1-\frac{d\Omega_{\rm DE}}{d\Omega_{\rm vis}}=-\frac{1}{4}+\frac{8}{9}\Omega_{\rm vis}+O\left(\Omega_{\rm vis}^2\right)$$

giving $d\Omega_{DE}/d\Omega_{vis}=-0.805654$ and $d\Omega_{DM}/d\Omega_{vis}=-0.194346$ at $\chi_{freeze}=3$ (-0.801715 and -0.198285 at $\Omega_{vis}=e^{-3}$), with open-valve equation of state strictly $w_{DE}=-1$ ($w_a=0$).

3. **Orthogonal 3-axis low-ℓ multipole cascade signature ($\ell=2,4$):** Because the child birth cascade proceeds through the sequential orthogonal handoffs $e_z\to e_x\to e_y$ with exact harmonic ratios $-1/2$ and -2 and multipole coupling $\partial_{q_z} g_{l,0}|_{q_z=0}=-(l-1)/(2)\partial_{q_1} g_{l,0}$ (giving $-(1)/(2)$ for the $\ell=2$ quadrupole and $-(3)/(2)$ for the $\ell=4$ hexadecapole), the surviving large-angle horizon imprint carries a deterministic planar quadrupole/octupole phase lock rather than statistically isotropic Gaussian random phases.

Proposition 9.7b (Condensed Baryon Core vs. WHIM Filament Split, $H_0=67.03$ km s^{-1}Mpc^{-1}, and Sub-Horizon S_8 Growth Suppression): Decomposing the visible and dark-matter budgets on

$9DVB=4(V+B/2)^3$ at $\chi_{freeze}=3$ into their localized vs. boundary/filament components yields three quantitative observational consequences:

- **(1) Condensed `(1,1,1)` baryon cores vs. uncondensed 1D WHIM filaments ($H_0=67.03$ km $s^{-1}Mpc^{-1}$):** The total visible stretch share at $\chi_{freeze}=3$ is $\Omega_{vis}(3)=5.3039\%$, whereas the 3D volumetric triad floor is $\Omega_{core}=V_{3D}(3)=e^{-3}=4.9787\%$. Their difference,

$$\Delta\Omega_{\mathrm{WHIM}} = \Omega_{\mathrm{vis}}(3) - e^{-3} = 5.3039\% - 4.9787\% = 0.3252\%$$
$$(6.13\% \text{ of } \Omega_{\mathrm{vis}})$$

identifies the uncondensed 1D filament stretch remaining in the warm-hot intergalactic medium (WHIM) outside localized `(1,1,1)` baryon triads. Identifying the condensed baryon share with $\Omega_b=e^{-3}=0.04978707$ and combining with the CMB acoustic baryon density $\Omega_b h^2=0.02237\pm 0.00015$ determines the dimensionless Hubble parameter without free parameters:

$$h = \sqrt{\Omega_b h^2\, e^3} = \sqrt{0.02237 \times 20.085537} = 0.67031 \pm 0.0022 \Longrightarrow H_0 = 67.03 \pm 0.22 \ \mathrm{km\,s^{-1}Mpc^{-1}}.$$

- **(2) Background vs. clustering dark-matter split and S_8 suppression:** In Theorem 1, the dark-matter share $M=\Omega_{DM}$ decomposes into the isotropic child `0`-horizon enclosing load $\Omega_{DM,throat}=(1)/(3)D=(\theta^*)^{3/2}/E^*$ and the radial infall load $B=M-(1)/(3)D=2(1-\theta^*)e^{-\chi}/E^*$ (supplemented on sub-horizon scales by the transmitted massive scalar dust tail $\langle\rho_\chi\rangle\propto a^{-3}$ of Section 5). Because the smooth horizon boundary load is spatially uniform across the bounding `0`-surface ($\nabla\rho_{DM,throat}=0$), it contributes fully to the background Friedmann expansion rate $H^2(a)=H_0^2[\Omega_{DE}+(\Omega_{DM}+\Omega_{vis})a^{-3}]$, whereas only the localized infall/scalar-dust fraction $f_{clust}=\Omega_{m,clust}/\Omega_m<1$ sources the sub-horizon Poisson equation on scales $k/(aH)\to\infty$. Sub-horizon matter density perturbations $\delta_m(a)$ therefore obey the modified Meszaros growth equation

$$\frac{d^2\delta_m}{d(\ln a)^2} + \left(2 + \frac{d\ln H}{d\ln a}\right)\frac{d\delta_m}{d\ln a} - \frac{3}{2} f_{\mathrm{clust}}\,\Omega_m(a)\,\delta_m = 0$$

which shifts the logarithmic growth rate by $f_{growth}(a)=d\ln\ \delta_m/d\ln\ a\approx\Omega_m(a)^{6/11}[1-(3)/(5)(1-f_{clust})]$. A smooth horizon boundary residual of $1-f_{clust}=1-\sqrt{\Omega_{vis}(\chi_*)/\Omega_{vis}(3)}=1-\sqrt{0.049787/0.053039}=3.114\%$ suppresses late-time linear clustering from the Planck CMB extrapolation $S_{8,CMB}\approx 0.832$ to

$$S_{8,\,\mathrm{NSC}} \approx S_{8,\,\mathrm{CMB}} \left(\frac{a_{\mathrm{eq}}}{a_0}\right)^{+\frac{3}{5}(1 - f_{\mathrm{clust}})\bar{\Omega}_m} \approx 0.776$$

directly reconciling early-universe CMB acoustic peaks with late-universe weak-lensing shear measurements ($S_8\in[0.76,0.79]$).

9.8 Harmonic asymmetry ladder, non-local-to-local observer dictionary, and strict quarantine of non-derived coincidences

Evaluating the half-cycle spectral asymmetry $\eta(\lambda)=\tanh((\pi)/(2)(\lambda)/(\lambda_{max}))$ along discrete 3D volumetric e-folds e^{-3K} anchored at the spaghettification threshold $e^{-2\chi_{spaghetti}}=1/2$,

$$\eta_K = \tanh\left(\frac{\pi}{2} e^{-2\chi_{\mathrm{spaghetti}}}\, e^{-3K}\right) = \tanh\left(\frac{\pi}{4} e^{-3K}\right)$$

gives $\eta_7=\tanh((\pi)/(4)e^{-21})=5.9553290339\times 10^{-10}$ at stage $K=7$ ($3K=21$ volumetric e-folds, i.e., 7 full 3-axis cascade cycles $7\times\chi_{freeze}=21$):

3D harmonic stage K	Volumetric e-folds $3K$	Scale factor e^{-3K}	Asymmetry $\eta_K=\tanh((\pi)/(4)e^{-3K})$	Regime
$K=1$	3	4.9787×10^{-2}	3.9083×10^{-2}	Macroscopic 3-axis cascade floor ($\chi_{freeze}=3$)
$K=3$	9	1.2341×10^{-4}	9.6926×10^{-5}	Intermediate vortex cascade
$K=5$	15	3.0590×10^{-7}	2.4026×10^{-7}	Pre-freeze-out cascade
$K=6$	18	1.5230×10^{-8}	1.1962×10^{-8}	Pre-baryonic harmonic
$K=7$	21	7.5826×10^{-10}	5.955329×10^{-10}	**BBN window** $[5.8,6.5]\times 10^{-10}$
$K=8$	24	3.7751×10^{-11}	2.9650×10^{-11}	Sub-baryonic harmonic

Complete Non-Local to Local Observer Dictionary:

Non-local multiverse geometry (A0--A4)	Local physics label	Derived mathematical relation & calculated readout
Covariant 3+1D Hamiltonian $\int_V (T^{00}_{phase}+V(s_*))\, d^3x$ & Σ_0 junction variation	**A0** functional & open-valve stationarity	Exact reduction $E=\sum_{i=1}^{3} m_i/\lambda_i+PV_0\lambda_1\lambda_2\lambda_3$ ($A_{\perp,i}=\pi/\mu^2(s_*)$); $\delta S_{\Sigma_0}/\delta\theta=0\Leftrightarrow\mu_+=\mu_-\Leftrightarrow\partial E/\partial\theta=0$ (Propositions 1.1--1.1b)
Circle $z(\tau)=R_0e^{i\omega\tau}$ seen edge-on as $R(\tau)=R_0\cos(\omega\tau)$ + hard core R_{core}	Black-hole singularity / Big Bang bounce stress	Smooth signed crossing ($\Delta R=0$, $\dot{R}(\tau_0)=0$) removes cusp $\Delta\dot{r}=+2R_0\omega$; $R=\sqrt{(R^2+R_{core}^2)}$ gives $R_4(0)=-\kappa/\pi$, $T_{\hat{a}\hat{b}}(0)=(1)/(2\pi)diag(-1,-1,+1,+1)$ (Propositions 1.2--1.2b)
Two half-turns $(-1)\times(-1)=+1$ across two nested `0` crossings ($P^2=(-I_3)^2=+I_3$)	Parity / orientation restoration across two generations	$\det P=(-1)^3=-1$ (*Parent (+)→Child (-)*), $\det(P^2)=(-1)(-1)=+1$ (*Child (-)→Grandchild (+)*)
Eliminating (θ^*,χ) from 5-variable budget shares (D,M,V) with $B=M-D/3$	Cosmological dark-sector / baryon partition curve & EoS	Exact cubic $9DVB=4(V+B/2)^3$, open-valve $dE_{DE}^*=(-(1)/(3)+(4)/(3))\rho_{DE}dV\Rightarrow w_{DE}=-1$, $w_{DM}=0$, $\lim_{\chi\to\infty}B/V^2=16/27$
Exponential-wall bypasses (`1D -> helix`, `+ -> 0 -> -`, single-axis $N^{3/2}\to$ 3D triad `(1,1,1)`)	3D nucleon ground state & nuclear hard-core potential	`(1,1,1)` ($N=3$) is the unique minimum-cost 3D ground state ($E^*=4.0$, *1:1:1* sphere); `(2,1,1)` ($N=4$) is *4:1:1* prolate ($E^*=4\sqrt{2}$); two `(1,1,1)` triads attract at $d>2R^*$ ($-2m_0/d$) and repel at $d\to 0$ against `(2,2,2)` ($+3.3137$ hard core)
Gauss-divergence exterior Virial $W_{bulk}:W_{throat}=3:1$ & loop closure $z(1)=z(0)$	Added mass in surrounding flow, quantized line bursts & $180°$ annihilation	$W_{flow}=-3T_{ext}-T_{ext}=-4T_{ext}$ (*75%* in $r>R$ bulk, *25%* at $r=R$ throat, *0%* in a point); $(+1,-1)\to 0$ releases two equal $180°$ ripples
Coupled finite-parent / receiver continuation & hybrid radial inertia $M(r)=1/(1+r^2/3)$	Wave-driven child opening & last-echo horizon cutoff	Zero-radius opening $r(\tau)=\pi^2\tau^2-(4)/(3)\pi^4\tau^3+O(\tau^4)$; last-echo cutoff $\tau_{echo}=0.87596883$ ($\Delta E_{post}/E_{tot}=0$, visible interior snapshot *25.63%*)
Whole-exterior causal neck onset & harmonic trace $\partial_{\zeta_x}v+\partial_{\zeta_y}v+\partial_{\zeta_z}v=0$	Neck pinch waveform & 3-axis orthogonal cascade $e_z\to e_x\to e_y$	Characteristic cubic $p^3-3p-3=0$ ($p_+=\varphi^{2/3}+\varphi^{-2/3}\approx 2.1038034$); $1\to 2\to 4\to 8$ lobe cascade with exact ratios *-1/2* and *-2* and B-independent *det* $M_{full}=0.14003084>0$
Quantized cascade step $\Delta\chi_k=n_k=1$ ($\chi_{freeze}=3$), WHIM split & sub-horizon growth	Cosmic freeze-out, $H_0=67.03$ km $s^{-1}Mpc^{-1}$ & $S_8\approx 0.776$	$\chi_{freeze}=\sum_{k=1}^{3} n_k=3$, $(\Omega_{DE},\Omega_{DM},\Omega_{vis})=(70.88\%,23.81\%,5.30\%)$, $\Omega_b=e^{-3}=4.98\%$, $\Delta\Omega_{WHIM}=0.33\%$, $d\Omega_{DE}/d\Omega_{vis}\approx -0.806$ (Theorem 7, Lemma 7.1, Proposition 9.7b)

Strict quarantine of candidate numerical coincidences not derived from A0--A4: To maintain strict separation between proved geometric theorems and post-hoc numerical comparisons, the

following expressions are explicitly quarantined from the deductive theorems: (i) Lenz's 1951 number $6\pi^5 \approx 1836.1181$ and its dressing $6\pi^5+e^{-3}/\sqrt{2} \approx 1836.1533$ (the discrete **A0+A1** binomial ladder gives $E_{3D}^{*}(3)/E_{1D}^{*}(1)=6$ without a derived π^5 continuum factor); (ii) the threshold n–p gap formula $2+(1)/(2)+e^{-3}/\sqrt{2} \approx 2.5352\ m_e$; (iii) selecting rung $K=7$ on the η_K ladder without an independent freeze-out equation beyond noting $3K=7\chi_{freeze}=21$ (especially since Theorem 2 proves three $O(1)$ geometric bypasses of e^{-3K} dilution); and (iv) CMB temperature projections using measured H_0 and G or an ad-hoc $-4/5$ exponent.

Appendix A. Technical atlas and reproducibility index

The main argument follows the outward geometric chain. Sections A.1--A.9 summarize the nine preserved technical engines, and Section A.10 provides the complete reproducibility table binding each section to its verified mathematical owners and numerical datasets. Complete historical chapters, equations, tables, and claim bindings are retained in paper/technical-atlas-record.md and in the archived `v0.26.0` foundation volume.

A.1. Generalized-harmonic evolution engine

The compact nonspherical Einstein/two-scalar generalized-harmonic (GH) implementation, exact frame derivatives, constraint diagnostics and local operator controls evolve the declared covariant action of Section 2. Its existence does not select a cosmic preparation or prove a child; the full technical development is retained in the GH owner.

A.2. Coordinate conditioning

Metric-dependent lapse feedback and proper-duration comparisons control coordinate conditioning without adding physical stress. They remove an imposed coordinate-time ceiling rather than an invariant singularity or a missing vacuum law, as documented in the GH owner.

A.3. Observer readouts

Proper-clock, phase and local stress readouts distinguish lapse/shift coordinates from receiver measurements. Their finite apparatus protocols and supplied initial state are preserved in the GH comparison record without identifying the receiver event with today's cosmological epoch.

A.4. Boson transport

Canonical projection and symplectic matter-probe transport propagate perturbations on the supplied parent geometry. Reference response norms are not energy fractions; incoming covariance and omitted metric fluctuations remain explicit in the boson owner.

A.5. Dense polynomial history

Retained DOP853 dense-polynomial history supplies source-compatible metric and field samples for probe evolution without replacing the nonlinear solution by sparse interpolation or extrapolation, as recorded in the history summary.

A.6. Schur memory kernel

Retarded finite-sector reduction derives the memory kernel from canonical block evolution. Its noise requires an incoming covariance, and initial cross-correlations remain physical data; a memory kernel alone is not a vacuum stress (boson owner).

A.7. Gravitating kick and shear

York-constrained momentum preparations and amplitude-independent shear/internal-coupling relations remain finite conditional results with explicit source profiles and vacuum-reference curvature in the kick owner.

A.8. Finite-amplitude resolution controls

Three-resolution controls at $\varepsilon=0.25$ preserve the nonspherical preparation and fixed proper endpoint with decreasing constraint and adjacent readout differences (comparison record). Separate $\varepsilon=1$ nonlinear runs are **under-resolved, not used for claims** and establish no trapped threshold.

A.9. Matter-work ledgers

Perturbative balances, passive nonlinear ledgers and finite controls retain pressure, geometry, measure and boundary contributions rather than identifying energy loss with vacuum production, with residuals recorded in the coupled-response and GH owners.

A.10. Reproducibility and verification index

All symbolic derivations, numerical solvers, and pre-registered evaluations in Sections 1--9 are executable in src/recursive_horizons/nsc_atlas_offset_generator.py and authenticated against the repository records indexed below:

Manuscript section	Mathematical owner	Authenticated numerical dataset
Section 1 (Offset generator, metric completion, axioms **A0--A4**, Propositions 1.1--1.2)	Offset-generator proof	Offset-generator summary
Sections 2--3 (Smooth `0` horizon, flux/area/handoff law)	Parent circulation owner	GH history summary
Sections 4--5 (Pre-registered test NSC-P006, receiver pressure, dust tail)	Offset thermodynamics owner	Flow forecast, Flow evaluation
Section 6 (Finite-amplitude existence, Hebey--Vaugon bound)	Finite-parent data owner	Finite-parent data summary
Section 6.1 (Internal work, continuation, one-mode trapped neck, Dirac mixing)	Valve formation owner	Parent release, Continuation, Surface check, Valve neck, GH neck, Spherical ledger, Conformal continuation, Local restart, Source gate, Dirac modes, Dirac onset, Dirac source
Sections 9.1--9.3 & 9.7--9.8 (Theorems 1--3 & 7, budget cubic, open-valve EoS, triad hard core, *3:1* Virial split, $\chi_{freeze}=3$)	Offset-generator owner	Non-local-to-local atlas summary
Section 9.4 (Theorem 4, coupled parent/receiver, zero-radius opening, last-echo cutoff)	Coupled continuation owner	Finite parent/receiver summary
Section 9.5 (Theorem 5, whole-exterior neck onset $p^3-3p-3=0$, multipole waveform)	Offset-generator owner	Neck waveform summary
Section 9.6 (Theorem 6, 3-axis cascade $e_z\to e_x\to e_y$, 2×2 kinetic handoff)	Offset-generator owner	Inherited daughter summary